West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Automobile Engineering [AE]

Part-II (3rd Semester)

Revised 2022

Proposed Syllabus of Strength of Materials

Course Code	AEPC 301
Course Title	Strength of Materials
Number of Credits and L-T-P	3 [L – 2, T – 1, P - 0]
Course Category	PC
Prerequisites	Engineering Mechanics

Course Objectives:

After completing this course, the students will be able

- To apply the concept of Simple Stresses and Strains in solving related problems.
- To applythe concept of Strain Energy in solving related problems
- To understandthe concept of Shear Force and Bending Moment and apply the concept in drawing SFDs and BMDs to related problems.
- To understand and applythe concept of Theory of Simple Bending and Deflection of Beams.
- To understand the concept of Shear Stress in non-uniform bending.
- To understand and apply the concept of Torsion in Shafts and Springs.
- To understand and apply the concept of Principal stresses at a point in biaxial stress case.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Simple Stresses and Strains:** ➢ Types of forces – Gradual/static, impact, suddenly applied; tensile, compressive etc. ➢ Stress (normal, shear, tensile, compressive etc), Strain (both normal and shear) and their nature. ➢ Mechanical properties of common engineering materials – Rigidity, elasticity, plasticity, stiffness, ductility, malleability, hardness, toughness. ➢ Study of stress-strain diagram for M.S. and C.I. specimens; Explanation of yielding, plasticity, ductility and toughness from stress-strain diagram. ➢ Concept and significance of factor of safety and working/allowable stress. ➢ Stress,strain and deflection values in bodies of uniform	**10**

	section and of composite section under the influence of normal forces. ➢ Thermal stresses in bodies of uniform section. ➢ Hoop stress and longitudinal stress in thin cylinder; concept and derivation of formula. Related simple numerical problems. **Strain Energy:** ➢ Derivation of strain energy for the following cases: i) Gradually applied load, ii) Suddenly applied load, iii) Impact/shock load. ➢ Strain energy or resilience, proof resilience and modulus of resilience. Related simple numerical problems.	
2	**Shear Force & Bending Moment Diagrams:** ➢ Types of beams with examples: a) Cantileverbeam, b) Simply supported beam, c) Over hanging beam, d) Continuous beam, e) Fixed beam. ➢ Types of Loads – Point and distributed; UDL and UVL ➢ Definition and explanation of shear force and bending moment;Calculation of shear force and bending moment and drawing the S.F and B.M. diagrams by the analyticalmethod only for the following cases: a) Cantilever with point loads, b) Cantilever with uniformly distributed load, c) Simply supported beam with point loads, d) Simply supported beam with UDL, e) Over hanging beam with point loads, at the centre and at free ends, f) Over hanging beam with UDL throughout, g) Combination of point and UDL for the above; Related numerical problems.	**8**
3	**Theory of Simple Bending and Deflection of Beams:** ➢ Explanation of terms: -Pure bending, Neutral layer, Neutral Axis of both beam and cross-section, Section Modulus, Moment of Resistance, Bending stress, Radius of curvature. ➢ Assumptions in theory of simple bending. ➢ Bending Equation $M/I = \sigma/Y = E/R$ with derivation. Stress distribution diagram. ➢ Problems involving calculations of bending stress, modulus of section and moment of resistance; Calculationof safe loads and safe span and dimensions of cross- section.	**8**

	➢ Beam sections of interest in automotive application: I-section, Channel section, box section; Properties of these sections that make those suitable for automotive applications and areas of application. ➢ Shear stress in bending: - Rise of shear stress in non-uniform bending. Shear stress equation $\tau = \frac{VQ}{Ib}$; No derivation; Explanation of individual terms. Shear stress variation in rectangular and circular cross section. (No numerical problems from this topic) ➢ Definition and explanation of deflection as applied to beams; Deflection formulae without proof for cantilever and simply supported beamswith point load and UDL only (Standard cases only) Related numerical problems.	
4	**Torsion in Shafts and Springs:** ➢ Definition and function of shaft. ➢ Calculation of polar M.I. for solid and hollow shafts. ➢ Assumptions in simple torsion. ➢ Derivation of the equation $T/J=\tau/R=G\phi/L$; ➢ Problems on design of shaft based on strength and rigidity; ➢ Use of propeller shaft, torsion bar inautomobiles. ➢ Numerical Problems related to comparisonof strength and weight of solid and hollow shafts; ➢ Classification of springs; Nomenclature of close coiled helical spring.Use of helical springs in automobiles. ➢ Shear stress and deflection formula for closed coil helical spring (without derivation).Stiffness of spring (No composite spring). ➢ Numerical problems on close coiled helical spring to find safe load, deflection, size of coil and number of coils.	**8**
5	**Principal Stress:** ➢ Stresses on an inclined plane. ➢ Complementary shear stress. Naming convention of shear stress. ➢ Ideal case of biaxial/plane stress; Values of normal and shear stresses on an inclined plane in case of biaxial stress. (No derivation – trigonometric expressions only) ➢ Concept of Principal planes,Principal stresses, Maximum shear stress; their values and positions. ➢ Mohr's circle of stress. Derivation of values of σ_1, σ_2 and	**8**

	τ_{max} from Mohr's circle. ➢ Construction of Mohr's circle in the following cases: Uniaxial stress (a rod/bar in pure tension/compression);pure shear (an element on a shaft in pure torsion); equal tension and compression on two perpendicular planes (no shear); mention equivalence of the previous two scenarios; Discuss stress scenarios on elements at various positions of a beam under non-uniform bending – on neutral axis, on two extreme fibres, somewhere in between neutral axis and extreme fibres. ➢ Numerical problems on Principal stress. ➢ Numerical problems on Mohr's circle. ➢ Theories of Failure: - Principal stress theory, Principal strain theory, Shear stress theory; Their applications and limitations.	
	Total Number of Contact Hours	**42**

Weightage distribution in both objective, short and long answer type questions:

Group Name	Module Number	Weightage (%)
A	1	20
B	2 & 3	40
C	4 & 5	40

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Compute stress and strain values and find the changes in axial dimension of bodies of uniform section under the influence of normal forces.
CO2	Calculate thermal stresses, in bodies of uniform section.
CO3	Define resilience, proof – resilience and modulus of resilience and obtain expressions for instantaneous stress developed in bodies subjected to different loads.
CO4	Compute shear force and bending moment at any section of beam and draw the S.F. & B.M diagrams and Calculate the safe load, safe span and dimensions of cross section of beams subjected to UDL and Point loads.
CO5	Compare strength and weight of solid and hollow shaftsof the same length and material subjected to torsionand compute the stress and deflection of the closed coil helical spring.
CO6	Compute maximum/minimum normal and shear stresses in case of biaxial/ plane stresses.

Text Books:

1. Strength of Materials – S.S. Bhavikatti, Vikas Publishers
2. Strength of Materials – U.C. Jindal, Pearson
3. Strength of Materials – R.S. Khurmi, S.Chand Company Ltd. Delhi
4. A Text Book strength of Material– R.K. Bansal, Laxmi Publication New Delhi
5. Strength of Materials – D.S. Bedi, Khanna Book Publishing Co. (P) Ltd., Delhi, 2017
6. Strength of Materials – B.C. Punmia, Ashok Kumar Jain & Arun Kumar Jain, Laxmi Publications,New Delhi, 2013

Reference Books:

1. Strength of Materials (SI Edition) – R.C Hibbeler, Pearson
2. Strength of Materials – Stephen Timoshenko, CBS
3. Strength of Materials – R. Subramanian, Oxford

Proposed Syllabus of Heat Power Engineering

Course Code	AEPC 302
Course Title	Heat Power Engineering
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	Nil

Course Objectives:

After completing this course, the students will be able

- To know the applications of thermodynamic principles and processes.
- To recognize the working of various Steam Boilers, functions of various accessories and mountings of boilers.
- To differentiate the different air standard cycle and calculate their efficiency
- To realize the working process of air compressor and a compression between signal stage and multi stage air compressor and different types of air compressor.
- To achieve some idea on refrigeration process and air conditioning process.
- To aware regarding different types of heat transfer process and their application

Module No.	Description of Topic	Contact Hrs.
1	**Basic of Engineering Thermodynamics** 1.1. Thermodynamics, System, Surroundings, Universe, 1.2. Types of Systems- closed system, open system, isolated system, flow system, non-flow system, and examples. 1.3. Properties of systems- Extensive and Intensive properties with symbols and units. Thermodynamic state, Path, Process, Quasi-static process, reversible, irreversible, cycle, thermodynamic equilibrium, Point function and path function. 1.4. Pressure- Units, standard atmospheric pressure, gauge pressure, Absolute pressure. Temperature- units, Meaning of N.T.P. and S.T.P. 1.5. Energy: —Types, Thermodynamic definition of Heat and Work, Difference between heat and work, internal energy, entropy, flow work and enthalpy. 1.6. Perfect gas and real gas, Equation of state, characteristic gas constant, Mole of gas, Universal gas constant.	**10**

	1.7. Specific heats of ideal gas, relation between specific heats and Characteristic gas constant. 1.8. Zeroth Law of Thermodynamics 1.9. First law of Thermodynamics, Mechanical equivalent of heat, General energy equations for non-flow process & flow process, principle of conservation energy. 1.10. Second law of Thermodynamics- statement (Kelvin-Planck & Claudius), C.O.P of refrigerator and heat pump. (Simple numerical). 1.11. Various Thermodynamic processes: Isobaric, isochoric, Isothermal, Adiabatic, Isentropic and Polytrophic with representation on P-V & T-S diagram, formula for work done, heat transfer, change in internal energy, enthalpy, entropy etc. (Derivations) (Simple numerical)	
2	**Air Standard cycles** 2.1. Introduction – assumptions – efficiency of the cycle, Reversible cycle – Irreversible cycle. 2.2. Types of air Standard thermodynamic cycles – Carnot cycle –Otto cycle – Diesel cycle – Dual combustion cycle,- Joule's cycle -Open and close Brayton cycle– derivation – P-V & T-S diagram, their efficiencies & simple problems.	**7**
3	**Properties of steam andmain components of steam power plant** 3.1. Formation of steam, Basic terms associated with steam formation: saturated liquid line, saturated vapour line, liquid region, vapour region, wet region, superheat region, critical point, saturated liquid, saturated vapour, saturation temperature, sensible heat, latent heat, wet steam, dryness fraction, wetness fraction, saturated steam, superheated steam, degree of superheat 3.2. Steam Table & its use, Enthalpy- Entropy diagram of steam (Mollier Chart) and its use. (Simple Numerical) 3.3. Determination of enthalpy, internal energy, internal latent heat, entropy of wet, dry and superheated steam at a given pressure and temperature using steam tables (Simple Numerical) 3.4. Study of boiler-Water Tube & Fire Tube boiler, boiler mounting and Accessories. [only name, location in boiler and function] 3.5. Steam condenser, function, location in steam power Plant. 3.6. Steam Turbine: Classification of turbine, location in steam	**6**

	power plant.	
4	**Air Compressors** 4.1. Uses of compressed air; 4.2. Functions and Types of air compressors; Single stage reciprocating air compressor - its construction and working (with line diagram) using P-V diagram; (simple Numerical) 4.3. Multi stage compressors – Advantages over single stage compressors, - inter cooler. (No numerical) 4.4. Rotary compressors: Centrifugal compressor, axial flow compressor, screw compressor and vane type compressors.	**4**
5	**Refrigeration & Air-conditioning** **<u>Refrigeration</u>** 5.1. Definition – Refrigeration, heat pump, heat engine, C.O.P, EER (Energy Efficient Ratio),ISEER (Indian Sessional Energy Efficient Ratio), BEE Star Rating,& unit of refrigeration 5.2. Refrigerant – properties required, common commercial refrigerants, Environment friendly refrigerants. 5.3. Air Refrigeration system, Bell Coleman cycle (reversed Joule cycle): components, working & applications; 5.4. Vapour Compression system: components, working & applications (schematic layout, p-h diagram, function & working of each component in the circuit)..(No Numerical) **<u>Air conditioning</u>** 5.5. Definition and classification of Air conditioning 5.6. Basic concept of Psychrometry including the following: 5.7. Dry air – Moist Air, - Water Vapour, - Saturated air – dry bulb temperature - wet bulb temperature – dew point temperature -humidity - specific and relative humidity and Degree of saturation. Psychrometer. 5.8. Psychometric chart and its uses – psychometric processes – sensible heating and cooling – humidification – dehumidification.(No Numerical) 5.9. Industrial Air-Conditioning; Window Air-Conditioner; Summer Air-Conditioning system, Winter Air-Conditioning system, Year-round Air-Conditioning system. 5.10. Car Air conditioning system. (only block diagram and working)	**9**

6	**Heat Transfer** 6.1. Modes of heat transfer--- Conduction, Convection and Radiation. 6.2. Conduction- Fourier's law, thermal conductivity- its units. Conduction through composite walls, Thermal resistance. [Simple numerical]. 6.3. Convective heat transfer – Newton's law of cooling, Heat Transfer Co-efficient – its units, Combined conduction and convection, Heat exchanger, types & application, Overall heat transfer co-efficient. [Simple numerical]. 6.4. Heat transfer by Radiation [for introductory concept only]: -- Thermal Radiation, Absorptivity, Transmissivity, Reflectivity, Emissivity, black and grey bodies, Stefan-Boltzman law. (No numerical) 6.5. Application of concept of heat transfer in automobiles.	**6**
	Total Number of Contact Hours	**42**

Weightage distribution in both objective, short and long answer type questions::

Group Name	Unit Number	Weightage (%)
A	1 & 2	40
B	3, 4 & 5	45
C	6	15

Course Outcome:

At the end of the course, the student will be able to:

CO1	Familiar with heat, work, thermodynamics law and different thermodynamics process.
CO2	Calculate the work done, enthalpy, internal energy, entropy and thermal efficiency of air stander cycle.
CO3	Distinguish between water tube and fire-tube boilers and explain the function all the mountings and accessories of boiler and turbine and also know the uses of steam tables.
CO4	Describe the constructional features of air compressor and working of different air compressors.
CO5	Know about the refrigeration and air-conditioning process and different psychometric properties.
CO6	Methodize the different heat transfer process and their application.

Text Books:

1. Thermal Engineering – R.S. Khurmi and J.K. Gupta, 18th Edition, S. Chand & Co, NewDelhi.
2. A Course in Thermal Engineering – S. Domkundwar& C.P. Kothandaraman, Dhanpat Rai& Publication, New Delhi

Reference Books:

1. Treatise on Heat Engineering in MKS and SI Units – V.P. Vasandani& D.S. Kumar, Metropolitan Book Co. Pvt. Ltd, New Delhi.
2. Thermal Engineering – P.L. Ballaney, Khanna Publishers, 2002
3. Thermal Engineering – R. K. Rajput,8th Edition, Laxmi publications Pvt Ltd, New Delhi.
4. Thermal Engineering - Mahesh M. Rathore, McGraw Hill India

Proposed Syllabus of Automotive Materials and Manufacturing Processes

Course Code	AEPC 303
Course Title	Automotive Materials And Manufacturing Processes
Number of Credits and L-T-P	42 (L-3, T-0, P-0)
Course Category	PC
Prerequisites	Nil

Course Objectives:

1. To gain knowledge about different categories of engineering materials along with their properties and application in automotive industries
2. To gain knowledge about different types of manufacturing processes along with their characteristics and proper use in automotive industries
3. To gain knowledge about different manufacturing processes those enhance the mechanical properties of the metallic materials required for automotive industries
4. To gather basic knowledge about formation of different standard surfaces by means of machine tools like lathe

CourseContents:

Module No.	Description of Topic	Contact Hrs.
1	**1. Engineering Materials** 1.1. Introduction 1.2. Classification of engineering materials 1.3. General properties, uses of different types of engineering materials	**(2)**
2	**2. Metals & Alloys** 2.1. Ferrous & nonferrous metals & its alloys used for automotive 2.2. Steels: carbon steel and alloy steels used in automobile industries 2.3. Effects of alloying elements on steel like chromium, nickel, silicon, molybdenum, tungsten, etc. 2.4. Application of various steel along with their composition 2.5. Aluminium & its alloys, such as wrought Aluminium, 2xxx series (alloyed with Copper), 3xxx series (alloyed with Manganege), 4xxx series (alloyed with Silicon) & 5xxx series (alloyed with Magnesium): composition, properties and use	**(4)**

	2.6. Copper & its alloys such as wrought copper, leaded copper, leaded brass, leaded bronze: composition, properties and use	
3	**3. Polymers, Ceramics & Composite Materials** 3.1. Thermoplastics and thermosetting plastics: properties and applications 3.2. Thermoplastics-Nylons and Polypropylene. Thermosetting Plastics-Epoxy resins and Polyesters Rubber – Natural andsynthetic: Properties & applications (no detail) 3.3. Properties and applications of different ceramic materials like glass, in automotive industries 3.4. Composite materials: Metals MatrixComposite, CeramicsMatrixComposite, Polymers MatrixComposite such as Carbon Fibre Reinforced Plastic, Carbon Fibre Reinforced Plastic, Natural Fibre Reinforced Plastic: Application area	**(4)**
4	**4. Manufacturing processes** 4.1. Definition and classification of manufacturing processes 4.2. Concepts of formative, subtractive and additive manufacturing; their applications 4.3. Concepts of property improvement processes and joining processes; their examples	**(2)**
5	**5. Formative manufacturing processes** 5.1. Different formative processes used for manufacturing automobile parts: casting, forging, rolling, extrusion 5.2. Working principle and use of forging, rolling and extrusion. 5.3. Classification of forging processes with application area 5.4. Examples of different automobile components made of forging, rolling, extrusion 5.5. Working principle and use of casting process 5.6. Concepts of moulding and moulding sand 5.7. Concepts of pattern and their allowances 5.8. Working principle, advantages and limitations of some modern casting processes like centrifugal casting, die casting and shell moulding 5.9. Concepts of some press works like blanking, piercing, notching, lancing, spinning, trimming, shaving and deburring.	**(12)**
6	**6. Subtractive manufacturing processes** 6.1. Chip and cutting tool 6.2. Mechanism of chip formation 6.3. Type of chip 6.4. Type of cutting tool:	**(4)**

	• single point and multi-point cutting tool • carbide tools, ceramics tools and CBN tools 6.5. Speed, feed, depth of cut both 6.6. Surfaces produced by different machine tools such as Lathe, drilling, milling, shaping machine	
7.	**7. External cylindrical surfaces manufacturing** 7.1. Lathe and its characterises 7.2. Specification of lathe 7.3. Important components of a Lathe and its function 7.4. Job holding devices, such as centre, chucks, dog, face plate, etc. 7.5. Tool holding devices 7.6. Types of tool used in lathe like solid shank, insert type, brazed tool, etc. 7.7. Different operation performed by a lathe to produce different external cylindrical surfaces like • Turning, • Taper turning • Thread cutting • Knurling • Grooving and shouldering 7.8. Cantering and facing operation done by lathe	**(7)**
8.	**8. Properties improvement processes: Heat treatment** 8.1. Heat treatment is a manufacturing process 8.2. Concept of phase and phase transformation 8.3. Iron-Iron Carbide (Fe-Fe3C) equilibrium diagram 8.4. Common heat treatment processes and their application • Annealing • Normalizing • Hardening, • Tempering 8.5. Principle and use of surface hardening processes like • Carburizing, • Nitriding, • Cyaniding, 8.6. Induction and flame hardening	**(7)**
	Total Number of Contact Hours	**42**

Weightage distribution in both objective, short and long answer type questions::

Group Name	Unit Number	Weightage (%)
A	1, 2 & 3	40
B	4, 5 & 6	40
C	7 & 8	20

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Illustrate different fundamental categories of engineering materials along with their properties and application in automotive industries
CO2	Explain fundamental classes of manufacturing processes such as formative manufacturing processes along with different sub-categories and their use in automotive industries
CO3	Explain fundamental classes of manufacturing processes such as subtractive manufacturing processes and their use in automotive industries
CO4	Explain working principles, characteristics and method of forming different types of external cylindrical surfaces produced by a Lathe

Reference Books:

1. A Text Book of Material Science & Metallurgy – O.P. Khanna, Dhanpath Rai and Sons, New Delhi. 2003.
2. Material Science & Engineering – R.K. Rajput, S.K. Kataria & Sons, New Delhi, 2004.
3. Material Science – R.S. Khurmi, S. Chand & Co. Ltd., New Delhi, 2005.
4. Manufacturing technology – P N Rao, Tata McGraw-Hill Publications
5. Elements of workshop Technology (Volume I & II) – S. K. Hajra Chaudary, Bose & Roy, Media Promoters and Publishers Limited.
5. Manufacturing Technology, Metal Cutting & Machine tools– P. N. Rao, Tata McGraw-Hill Publications
6. Production Technology – R.B. Gupta, Satya Prakashan, New Delhi

Proposed Syllabus of Automotive Chassis

Course Code	AEPC 304
Course Title	Automotive Chassis
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	NIL

Course Objectives:

After completing this course, the students will be able

1. To learn about the constructional details of chassis.
2. To gain knowledge about various steering system, steering linkages and steering gear boxes
3. To learn about the phenomenon of different components used in suspension and brake systems.
4. To learn about the various aspects of wheels and tyres.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Vehicle layout and Chassis frame** 1.1 Definition of an automobile. 1.2 General Vehicle layout; types of layout. 1.3 Classification of Vehicle layout with respect to Drive, Location of Engine, Arrangement of Engine (with Sketch) 1.4 Major assemblies – their locations and their functions. 1.5 Necessity of Frame and its functions 1.6 Type of frames, Conventional (Ladder and X – Member type), Semi integral and Integral types, frames construction, material, frame alignment. Frame sections- Channel, Box and Tubular Sections 1.7 Classification and Specifications of Chassis. 1.8 Defects in frames 1.9 Different Chassis Parameters (definitions)	**6**
2	**Front Axle and Steering** 2.1 Front Axle: Types of front axle - Dead axle, live axle. 2.2 Type of stub axle arrangements- Elliot, reverse Elliot,	**6**

	Lamoine, and reverse Lamoine. 2.3 Front wheel assembly. 2.4 Steering system. 2.5 Steering linkages. Steering geometry and its effects –Caster, camber and king pin Inclination, toe in– toe out, correct steering angle. Under steering and over steering, Turning radius. 2.6 Working Principle, Construction and application of Steering gear box – rack and pinion type, recirculating ball type, worm and roller type. 2.7 Construction and application of Steering gear box – worm and sector, screw and nut, worm and peg. 2.8 Adjustable steering system, Collapsible steering column, construction & working Principle. 2.9 Ackerman and Davis Steering Principle and linkage. 2.10 Defects & Troubleshooting	
3	**Suspension Systems** 3.1 Functions of suspension system. Sprung weight, unsprung weight, common suspension terms 3.2 Front and rear axle suspension, rigid and independent suspension, comparison. 3.3 Types of Rigid Axle suspension system - parallel leaf spring type, leading arm and trailing arm type, trailing arm with twist beam type, four link type 3.4 Types of Independent suspension system – McPherson Strut, Wishbone type 3.5 Leaf spring and their types, Helper springs (variable rate springs), coil spring torsion bar arrangement and shock absorber. 3.6 Use of Anti roll bar, stabilizer bar. 3.7 Shock absorbers – Telescopic and Gas Filled 3.8 Suspension system trouble shooting	**8**
4	**Brake System** 4.1 Introduction, Principle of braking. 4.2 Function and necessity of brakes, Braking Effect- weight transfer. 4.3 Classification of brakes and braking systems. 4.4 Construction and working of - Drum brake. 4.5 Concept of Leading Shoe & Trailing Shoe. 4.6 Friction materials used for brake shoes and pads. Characteristics of friction material - brake fade, coefficient	**14**

	of friction, dry friction and wet friction. 4.7 Disc brake, types, construction and working principle, use. 4.8 Solid and ventilated disc brakes. 4.9 Comparison between Drum Brake and Disc Brakes 4.10 Construction and working of - Mechanical braking system. 4.11 Hydraulic Braking system, construction & working principle. 4.12 Master cylinder, wheel cylinder construction & working principle, bleeding of Brakes. 4.13 Properties of Brake Fluid and their specifications, Brake Bleeding. 4.14 Exhaust brake, Parking brake, Adjustments of Brakes, Brake Test. 4.15 Brake System Troubleshooting.	
5	**Wheels and Tyres** 5.1 Wheels – Functions, Types of wheels, wired spoke wheel, disc and alloy wheels 5.2 Tyre- Necessity of Tyre, construction, working and comparison of a tubed tyre and tubeless tyre 5.3 Type of Rims. 5.4 Types of tyre- Radial, Cross Ply, Belted Bias Tyre. 5.5 Specification of tyre, Concept of Aspect Ratio 5.6 Types of Tread patterns 5.7 Effect of Inflation pressure on the life of tyre and tyre rotation 5.8 Tyre materials, construction, Tubular tyres. 5.9 Airless tyres 5.10 Balancing of wheel tyre assembling (static and dynamic). • Tyre Rotation. • Tyre Retreading. • Tube Vulcanizing. 5.11 Factors affecting tyre performance and life. 5.12 Troubleshooting of Wheels and tyres	**8**
	Total Number of Contact Hours	**42**

Weightage distribution in both objective, short and long answer type questions::

Group Name	Unit Number	Weightage (%)
A	1 & 2	40

B	3 & 4	40
C	5	20

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Classify the chassis layout with reference to the power train location.
CO2	Explain the different designs consideration of front axles and steering system for proper rolling of the tyre.
CO3	Explain the types of wheels and its constructional details which are used in Automobile.
CO4	Explain the various Braking systems and in which circumstances each one of them is used.

Text Books:

1. Automobile Engineering, Vol.1, Dr. Kripal Singh, Standard Publishers
2. Automotive Chassis, A. K. Babu, Scitech Publications

Reference Books:

1. Automobile Engineering, Prof. R.B. Gupta, Dhanpat Rai Publishing Company (P) Ltd.
2. A Textbook of Automobile Engineering, R.K. Rajput, Laxmi Publications
3. Automobile Engineering, Sudhir Kumar Saxena, Laxmi Publications
4. Automobile Engineering, Kamaraju Ramakrishna, PHI Learning Private Limited
5. Automobile Engineering, K. K. Ramalingam, Scitech Publications

Proposed Syllabus of Automotive Engines

Course Code	AEPC 305
Course Title	Automotive Engines
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	Nil

Course Objectives:

After completing this course, the students will be able

1. To gain knowledge on basics of automotive SI and CI engines, their types, construction, working principle and comparison.
2. To study the cooling and lubricating systems adopted in automobile engines.
3. To learn the properties of gasoline and diesel fuel and combustion process involved in petrol and diesel engines.
4. To gain knowledge on the fuel feed systems of petrol and diesel engines.
5. To study the different parameters relating to engine performance and to learn the pollution characteristics of SI and CI engine.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Engine Fundamentals and Constructional Features** 1.1 Engine terminology – bore, stroke, TDC, BDC, clearance volume, swept volume, compression ratio, volumetric efficiency. 1.2 Classification of engine, Concept and working of 4-stroke petrol engine, 4-stroke diesel engine, 2-stroke petrol engine, Comparison. 1.3 Engine working cycles – Theoretical and actual Otto & Diesel cycle. 1.4 Constructional features of engine components – (use, location, application, material) • Cylinder block and cylinder liners. • Piston assembly • Connecting rod • Crank shaft • Flywheel	**7**

	• Camshaft • Valve – Poppet valve, valve actuating mechanism, valve timing diagram • Manifolds – inlet and exhaust • Mufflers	
2	**Engine Cooling and Lubrication System** 2.1 Engine Cooling System – Purpose of cooling, types of cooling system. 2.2 Air cooling system – components, uses, merits and demerits. 2.3 Water cooling system components – water jacket, cooling pump, thermostat valve, cooling fan, radiator - working principle. 2.4 Engine Lubrication System – Function, types of lubricating system, properties of lubricating oil, rating of lubricating oil. 2.5 Layout and working of – Splash lubrication, petrol-oil lubrication, wet sump lubrication, dry sump lubrication, applications. 2.6 Causes of oil consumption and oil contamination, crank-case ventilation.	**8**
3	**Fuels - Properties & Combustion Theory** 3.1 Hydrocarbon fuels – Calorific value of fuel (HCV & LCV), Properties of SI and CI engine fuel, Octane Number, Cetane Number, Fire Point, Flash Point, Cloud Point. 3.2 Combustion in SI engine – Stages of combustion, Ignition lag, Flame propagation, Rate of pressure rise, Factors affecting combustion in SI engine. 3.3 Abnormal combustion in SI engine – Pre-ignition, Detonation, Surface ignition, their effects on performance of engine, control of abnormal combustion. 3.4 Combustion in CI engine – Stages of combustion, Delay period – its' significance, factors affecting delay period, importance of air motion in CI engine, Complete Combustion – importance of excess air. 3.5 Abnormal combustion in CI engine – Knocking – its' effects and control, difference between abnormal combustion in SI and CI engine. 3.6 Engine Troubleshooting - Probable causes and remedies of different symptoms like black smoke, blue smoke, white smoke, engine overheating, excessive fuel consumption, excessive oil consumption, different noises etc.	**8**

4	**Fuel Feed System in Petrol Engine** 4.1 Layout of fuel feed system in petrol engine, Gravity feed system. 4.2 Components of fuel feed system – Construction & Working of fuel pump (Mechanical & Electrical), air cleaner (dry & wet type), Simple Carburettor. Electronic Carburettor – Principle and application. 4.3 Air-fuel ratio – Concept of rich and lean mixture, Requirement of mixture strength at during different operating conditions like starting, idling, cruising, acceleration etc., Limitations of carburettor. 4.4 Petrol injection – Necessity, Advantages of petrol injection, Disadvantages, Comparison with carburetted fuel supply system.	**6**
5	**Fuel Feed System in Diesel Engine** 5.1 Layout of fuel feed system in diesel engines, Types of diesel injection system – Layout of Unit injection, Distributor system and Common Rail system. 5.2 Basic components – Fuel tank, Fuel lines, Fuel filters, Fuel feed pump, Air Cleaner, Fuel injection pump, Fuel injectors. 5.3 Fuel Injection Pump – Construction & operation of Inline and Distributor type pump, Phasing and calibration of FIP. Overview of electronic Fuel Injector. 5.4 Governor – Purpose of governor, Types of governor, Working of mechanical governor.	**6**
6	**Engine Performance & Testing** 6.1 Engine Performance parameters - Engine torque, Brake power, Indicated power, Frictional power, Mean effective pressure [on IP & BP], Fuel Consumption, Specific fuel consumption. 6.2 Engine efficiency – Indicated thermal efficiency, Brake thermal efficiency, Mechanical Efficiency, Efficiency ratio, Air standard efficiency, Volumetric Efficiency [on mass & volume]. 6.3 Measurement of brake power – Dynamometer, Types of dynamometer, Rope brake dynamometer. 6.4 Measurement of power of single cylinder and multi cylinder engines with the help of Morse test. 6.5 Making a heat balance sheet. 6.6 Simple problems to calculate engine power, efficiency etc.	**7**

	and preparation of heat balance sheet.	
	Total Number of Contact Hours	**42**

Weightage distribution in both objective, short and long answer type questions::

Group Name	Unit Number	Weightage (%)
A	1 & 2	30
B	3 & 4	40
C	5 & 6	30

Course Outcomes:

At the end of the course, the student will be able:

CO1	To understand the actual engine working principle of 2-stroke/4-stroke, SI and CI engines and their comparisons.
CO2	To define and identify various components of SI and CI engines and their sub-systems - cooling, lubrication and fuel feed system.
CO3	To gain basic knowledge on SI and CI engine combustion and its related parameters.
CO4	To apply their knowledge in analyzing the engine performance and pollution characteristics.

Reference Books:

1. R. B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi.
2. Dr. Kirpal Singh, Automobile Engineering (Vol. – 2), Standard Publishers Distributors, New Delhi.
3. S. Srinivasan, Automotive Mechanics, Tata McGraw Hill Education, New Delhi.
4. M. L. Mathur and R.P.Sharma, Internal Combustion Engine, Dhanpath Rai Publications (P) Ltd, New Delhi.
5. V. Ganesan, Internal Combustion Engines, Tata-McGraw Hill Publishing Co., New Delhi.

Proposed Syllabus of Heat Power Engineering Lab

Course Code	AEPC 306
Course Title	Heat Power Engineering Laboratory
Number of Credits and L-T-P	2 [L – 0, T – 0, P - 2]
Course Category	PC
Prerequisites	Nil

Course Objectives:

After completing this course, the students will be able

- To familiar with theworking of boilers and identify the differentaccessories and function of accessories.
- To know the process of dismantling and assembling of compressors and identify the different component.
- To differentiate the different type of heat exchanger.
- To identify the different component of Refrigeration and Air conditioning Unit.
- To differentiate between Refrigeration unit and Air Conditioning Unit

At least 06 practical have to be done.

Sl No.	Topics for Practice	Contact Hrs.	Remark (whether mandatory or not)
1	Study and compare various heat exchangers such as radiators, evaporators, condensers, plate type heat exchangers etc. by using either actual cut-section model or with the help of presentations/charts/animated videos/ wooden or metallic models etc.	6	Mandatory
2	Conduct performance test on A/C test rig to determine COP	6	At-list one from Sl. No 2 & Sl. No. 3
3	Conduct performance test on Refrigeration Test Rig to determine COP of the refrigerator	6	
4	Thermal conductivity test of a solid metallic rod or thick slab	4	At-list one from Sl. No 4 & Sl.No. 5
5	Calculation of STEFAN BOLTZMANN CONSTANT of sphere.	4	
6	Study of high-pressure boiler (With the help of presentations/charts/animated videos/ wooden or metallic models etc.)	4	Optional

7	Study of boiler mountings and accessories (With the help of presentations/charts/animated videos/ wooden or metallic models etc.)	4	Optional
8	Study of Refrigeration unit. (With the help of presentations/charts/animated videos/ wooden or metallic models/actuals cut section model etc.)	4	Optional
9	Study of Air Conditioning Unit. (With the help of presentations/charts/animated videos/ wooden or metallic models/ actuals cut section model etc.)	4	Optional
10	Dismantling and assembling of one reciprocating or one rotary compressor.	4	Optional
11	Conduct performance test on single stage reciprocating compressor	4	Optional
12	Demonstration of Leak detection of refrigeration unit. (by using either actual model or with the help of presentations/charts/animated videos/ wooden or metallic models etc	4	Optional
	Total Number of Contact Hours	**28**	

Course Outcome:

At the end of the course, the student will be able to:

CO1	Recognize the different component of compressor and boiler
CO2	Estimate the performance characteristics single stage air compressor
CO3	Find the thermal conductivity of metallic rod
CO4	Calculate the co efficient of performance of refrigerator and A/C
CO5	Characterize the different type of heat exchanger.

Reference Books:

1. Thermal Engineering – R.S. Khurmi and J.K. Gupta, 18th Edition, S. Chand & Co, New Delhi.
2. A Course in Thermal Engineering – S. Domkundwar & C.P. Kothandaraman, Dhanpat Rai& Publication, New Delhi

Proposed Syllabus of Automotive Materials and Manufacturing Processes Lab

Course Code	AEPC 307
Course Title	Automotive Materials and Manufacturing Processes Laboratory
Number of Credits and L-T-P	2 [L - 0, T - 0, P - 2]
Course Category	PC
Prerequisites	Nil

Course Objectives:

After completing this course, the students will be able

1. To gather knowledge about establishment of safety in working condition.
2. To gain knowledge about basic engineering materials used in automotive industries.
3. To know the different methods adopted in primary manufacturing processes like rolling, extrusion, forging, casting.
4. To develop knowledge and skill for promoting different metal cutting processes required in automotive industries made by lathe and drill along with press work.
5. To acquire knowledge about fundamental heat treatment processes adopted for enhancing mechanical properties of automobile parts.

CourseContents:

Sl. No.	Topics for Practice	Contact Hrs.	Remark (whether mandatory or not)
1	Develop one pattern for a given job considering all aspects of pattern making forgroup of 4 to 6 student. Job shall involve spit pattern with core, coreprint.	4	Mandatory
2	Prepare a sand mould for any one of the above patterns. Demonstrate the use of hand tools in preparation of moulding process.	5	Mandatory
3	Making of one job for group of 4 to 6 student involving followingoperations on a lathe machine: Facing, taper turning, step turning, threading, knurling operations **And/ or** Making of one job for group of 4 to 6 student involving followingoperations using a drilling machine: Drilling, reaming, counter boring, counter sinking, Spot	10	Mandatory

	facing operations.		
4	Prepare one wooden solid pattern per student as per givendrawing.	3	Optional (Any three of optional are compulsory)
5	With the help of presentations/charts/animated videos/wooden or metallic models etc. (wherever available) study of different Rolling processes along with their applications particularly in automotive field, advantages, limitations and probable defects.	3	Optional (Any three of optional are compulsory)
6	With the help of presentations/charts/animated videos/wooden or metallic models etc. (wherever available) study of Different Extrusion processes along with their applications particularly in automotive field, advantages, limitations and probable defects.	3	Optional (Any three of optional are compulsory)
7	With the help of presentations/charts/animated videos/wooden or metallic models etc. (wherever available) study of Different Forging processes along with their applications particularly in automotive field, advantages, limitations and probable defects.	3	Optional (Any three of optional are compulsory)
8	With the help of presentations/charts/animated videos/wooden or metallic models etc. (wherever available) study of Different Press Work processes along with their applications particularly in automotive field, advantages, limitations and probable defects.	3	Optional (Any three of optional are compulsory)
9	With the help of presentations/charts/animated videos/wooden or metallic models etc. (wherever available) study of different Heat treatment processes along with their typical applications in automotive industries, advantages and limitations.	3	Optional (Any three of optional are compulsory)
	Total Number of Contact Hours	**28**	

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Explain working principle, applications, advantages and limitations of different primary production systems along with press working.
CO2	Construct single piece pattern by individual students starting from a given drawing.
CO3	Develop pattern and sand mould projects in a group.

CO4	Demonstratedifferent fundamental machining operations on a lathe and/or drill machine.
CO5	Explain working principle, applications, advantages and limitations of heat treatment processes.

Reference Books:

1. S. K. Hajra Choudhury, A. K. Hajra Choudhury, Elements of Workshop Technology. Vol. - I and II, Media Promoters and Publishers Pvt. Ltd.
2. P. N. Rao, Manufacturing Technology, Vol. –I, II & III, The McGraw Hill Co.
3. Gerling, All about Machine Tools, Wiley Eastern Limited.
4. B. S. Raghuwanshi, Workshop Technology, Vol- I, II & III, Dhanpat Rai& Co.

Proposed Syllabus of Automotive Chassis Lab

Course Code	AEPC 308
Course Title	Automotive Chassis Laboratory
Number of Credits and L-T-P	2 [L – 0, T – 0, P - 2]
Course Category	PC
Prerequisites	Nil

Course Objectives:

After completing this course, the students will be able

1. To understand the various dimension and terms related to chassis.
2. To understand the working of steering system and steering gearbox.
3. To observe various parts of axles and understand their functions.
4. To perform various tests on wheel geometry and calculate the different parameters.
5. To understand various parts of braking and suspension system of a vehicle.

Course Content:

Sl. No.	Topics for Practice	Contact Hrs.	Remarks (whether mandatory or not)
1	Study of safety precautions to be followed and knowledge of first aid in an automobile workshop.	2	Mandatory
2	Study, measurement and drawing of schematic diagram of a car/ vehicle chassis. Parameters to be measured i. Wheel Base ii. Front Track iii. Rear Track iv. Ground Clearance v. Overall Length vi. Overall Height from Ground vii. Front Overhang viii. Rear Overhang ix. Angle of Approach x. Angle of Departure xi. Ramp break over angle	2	Mandatory (at least six parameters to be measured)

	xii. Eyebrow height (Front and rear)		
3	Study of steering system assembly.	2	Mandatory
4	Dismantling and assembling of steering gearbox. a) Rack and Pinion b) Recirculating Ball c) Worm and Roller	6	Mandatory
5	Study of different types of front and rear axles. i. Front Axle a) Dead axle b) Live axle ii. Rear Axle a) Semi floating rear axle b) Full floating rear axle c) Three quarter floating rear axle. iii. Stub Axle a) Elliot b) Reverse Elliot c) Lamoine d) Reverse Lamoine	6	Mandatory (any two stub axle)
6	Study and measurement of different wheel geometry. Parameters to be measured i. Caster ii. Camber iii. King pin inclination iv. Toe in– Toe out	2	Mandatory
7	Study and measurement of different parts of braking system. a) Disc Brake b) Drum Brake Parameters to be measured i. Master cylinder push rod length ii. Diameter of push rod iii. Travel of push rod (stroke of master cylinder) iv. Thickness of Pad for Disc v. Rotor thickness	4	Mandatory

	vi. Rotor swept area vii. Type of caliper viii. No of Piston (action and reaction type) ix. Caliper pin diameter x. Width of the shoe xi. Diameter of the drum xii. Shoe length xiii. Drum to shoe clearance xiv.Thickness of lining xv. Wheel cylinder (No of Piston) xvi.Type of shoe (L/T)		
8	Study and measurement of different parts of front suspension and rear suspension. a) McPherson Strut b) Wishbone arm c) Leaf Spring	4	Mandatory
	Total Number of Contact Hours	**28**	

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Evaluate the different dimensions related to chassis.
CO2	Describe the steering system, sub-systems and its application.
CO3	Distinguish between different types of axles and their usage.
CO4	Demonstrate the different wheel geometries.
CO5	Explain the application of different braking and suspension systems of automobile.

Reference Books:

1. Automobile Engineering, Prof. R.B. Gupta, Dhanpat Rai Publishing Company (P) Ltd.
2. A Textbook of Automobile Engineering, R.K. Rajput, Laxmi Publications

Proposed Syllabus of Automotive Engine Lab

Course Code	AEPC 309
Course Title	Automotive Engine Laboratory
Number of Credits and L-T-P	2 [L – 0, T – 0, P - 2]
Course Category	PC
Prerequisites	Nil

Course Objectives:

After completing this course, the students will be able

1. To know the use of different tools and special tools used for dismantling and assembling automobile engines.
2. To gather knowledge about the basic operation of different types of engines used in automobiles.
3. To identify the various components of the cooling and lubrication system and know their locations and functions.
4. To gain knowledge on the fuel feed systems of petrol and diesel engines.
5. To perform engine performance tests to calculate engine power and efficiency.

Course Contents:

Sl. No.	Topics for Practice	Contact Hrs.	Remark (whether mandatory or not)
1	Operate Cut Section model to know engine nomenclature, understand basic engine operation, identify, locate and compare various components of - a) Four stroke Petrol Engine b) Four stroke Diesel Engine c) Two stroke petrol Engine	8	Mandatory
2	Conduct engine performance test in Engine Test Rig for petrol/ diesel engines to find engine performance parameters and prepare heat balance sheet.	6	Mandatory
3	Dismantle & Assemble an Engine and practice the same. Identify and know the operations and functions of different general & special tools used for dismantling and assembling the engine.	6	Mandatory
4	With the help of cut section models / charts / presentations / animated videos etc., study the cooling system of an engine; identify and locate the	4	Optional

	various components and draw its layout.		
5	With the help of cut section models / charts / presentations / animated videos etc., study the lubricating system of an engine; identify and locate the various components and draw its layout.	4	Optional
6	With the help of cut section models / charts / presentations / animated videos etc., study the various components of the fuel supply system of petrol & diesel engines, compare their operation and draw their layout.	4	Optional
7	Construct the Valve timing diagram of a four-stroke engine.	4	Optional
	Total Number of Contact Hours	**28**	

Note: Students must perform six practical with Sl. No. 1, 2 and 3 as compulsory and any two from the optional.

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Explain the operation of different engines used in automobiles, identify and locate their components.
CO2	Describe the operation, location and function of the components of cooling and lubricating system of automobiles.
CO3	Demonstrate the location and function of the components of the fuel supply system.
CO4	Conduct engine performance test to find out the power, efficiency and other performance parameters of an engine.

Reference Books:

1. R. B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi.
2. Dr. Kirpal Singh, Automobile Engineering (Vol. – 2), Standard Publishers Distributors, New Delhi.

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Automobile Engineering [AE]

Part-II (4th Semester)

Revised 2022

Semester-wise Detailed Curriculum

Semester IV (Second year)

Branch/Course: Automobile Engineering

Sl. No.	Category	Code	Course Title	Hours per week			Total contact hours/ week	Credits
				Lecture	Tutorial	Practical		
1	Program Core Course	AEPC401	Automotive Powertrain	3	0	0	3	3
2	Program Core Course	AEPC402	Fluid Mechanics & Machines	3	0	0	3	3
3	Program Core Courses	AEPC403	Automotive Component Design	2	1	0	3	3
4	Program Core Courses	AEPC404	Theory of Machines & Mechanism	3	0	0	3	3
5	Program Elective Course	AEPE411 or AEPC412	Any one Programme Elective	3	0	0	3	3
6	Program Core Course	AEPC405	Automotive Powertrain Lab	0	0	2	2	1
7	Program Core Course	AEPC406	Strength of Material and Fluid Mechanics Lab	0	0	3	3	1.5
8	Program Core Course	AEPC407	Automobile Engineering Drawing and Computer Graphics Lab	0	0	4	4	2
9	Minor Project	PR202		0	0	4	4	2
Total				**14**	**1**	**13**	**28**	**21.5**

List of Programme Electives for Fourth Semester

1. AEPE 411: Automotive Pollution & Control
2. AEPE 412: Alternate Fuels and Energy Systems

Proposed Syllabus of Automotive Powertrain

Course Code	AEPC401
	Automotive Powertrain
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	NIL

Course Objectives:

1. To impart knowledge of various components of the Automotive powertrain.
2. To impart knowledge concerned to the power transmission phenomenon and improving the performance of vehicles.
3. To have a clear understanding of the different types of modern powertrain systems adopted in automobile vehicle.
4. To learn the different types of axle joints used in automobile vehicle.
5. To have a better understanding of final drive and its type.

Course Contents:

Module No.		**Description of Topic**	**Contact Hrs.**
1		**Automobile Clutches:** 1.1 Introduction, necessity, function and requirements of automotive Clutch. 1.2 Types of Automotive Clutch. - Friction and Non friction type Clutches. 1.3 Construction and Operation of Different type of clutches : 1.3.1 Construction and Operation of a single plate (coil and Diaphragm) dry disc clutch, multi plate, wet clutch, Centrifugal and Semi-Centrifugal Clutch. 1.3.2 Electromagnetic Clutch [Concept Only with the help of e-Sources] 1.4 Construction details of Clutch plate. Clutch lining materials, Pressure Springs, Torsional Springs. 1.5 Clutch Linkage, Clutch Adjustments. Self-Adjusting Clutch& Clutch Free Pedal Play. 1.6 Clutch operating mechanisms- Mechanical, Hydraulic &Vacuum. 1.7 Troubleshooting of all type of automotive clutch.	**08**
2	2.1	**Transmissions and Transaxles [Manual]:** 2.1.1 Purpose of the Transmission/ Transaxle - Need of Multiple Gear ratio and Multiple Torque setting. 2.1.2 Manual Transmissions and Transaxles, its' difference. 2.1.3 Function and types of [Transmission] Gear Boxes- Sliding Mesh, Constant Mesh, Synchromesh gear box -Construction, operation of each type, Power flow diagram and comparison among them. 2.1.4 Function and types of Transmission – Manual, iMT, AMT, DCT, CVT [Concept Only with the help of e-Sources] 2.1.5 Forward and Reverse Gear Ratio of different vehicles 2.1.6 Gear selector mechanism with gear lever on top of gearbox. 2.1.7 Transfer case, function, construction & power flow layout. 2.1.8 Function and types of Propulsion system – FWD, RWD, 4X4, AWD [Concept Only with the help of e-Sources]	**08**

		2.1.9 Lubrication of gear box. 2.1.10Troubleshooting of Gear box.	
	2.2	**Transmissions and Transaxles [Automatic]:** 2.2.1 Elements of Automatic Transmission. 2.2.2 Principle of Epi-cyclic Gearing and Gearing Ratio 2.2.3 Function, Construction & Working of Epicyclic Gear box. 2.2.4 Hydro-static drive and Hydro-dynamic drive 2.2.5 Fluid Coupling-Fluid-Flywheel-Torque Converter- Construction and working and application. 2.2.5 Freewheel Mechanism (Overrunning Clutch),Overdrive Mechanism. 2.2.6 Semi-Automatic Transmission, Control System. 2.2.7 Comparison with Conventional Transition System.	**08**
3		**Universal Joints and Propeller shaft:** 3.1 Necessity of Universal Joints. 3.2 Functions of universal joint and slip joint. 3.3 Types of Universal Joints, Constructional details of Universal Joint, Limitation of Universal Joint. 3.4 Necessity of CV joints. 3.5 Constant Velocity Rezappa and Tripod Joint. 3.5 Necessity of Propeller shaft. 3.6 Function and constructional features of Propeller Shaft. 3.7 Whirling of shaft, Two-piece Propeller Shaft. 3.8 Troubleshooting of different type of joints and Propeller Shaft.	**06**
4		**Final Drive and Differential:** 4.1 Necessity of Final Drive. 4.2 Types of Gears used for Final Drive & their comparison. 4.3 Final Drive Ratio & Overall Gear Ratio. Final drive ratio of different types of vehicles. 4.4 Differential - Necessity of Differential. 4.5 Construction and working of differential. 4.6 Differential lock & Differential Slip. 4.7 Function and types of Differential – Open Differential (OD), Locking Differential (LD), Limited Slip Differential (LSD), Torque Vectoring Differential (TVD)[Concept Only with the help of e-Sources] 4.7 Troubleshooting of Differential	**08**
5		**Rear Axle & Rear Axle Drive:** 5.1 Necessity of Rear Axle. 5.2 Loads acting on the rear axles. 5.3 Construction & working of different type of rear axle - Semi floating, Three quarter floating and Full floating type. 5.4 Rear Axle Drive –Types, construction and working principle of Hotchkiss Drive & Torque Tube Drive, its' use.	**04**
Total Hours			**42**

Weightage distribution in both objective, short and broad answer type questions:

Group	Module Number	Weightage (%)
A	1 & 2	40
B	3	30
C	4 & 5	30

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Analyze the principle, construction and working of elements of automotive powertrain system.
CO2	Distinguish between construction and working of various types of clutches.
CO3	Identify construction and working principle of various types of Gear Boxes.
CO4	Understand construction and principle of working of various types of Powertrain system used in modern automotive vehicle.
CO5	Analyze the working of final drive and differential action.
CO6	Understand construction and principle of working of various types of Shafts and axle.

Text Books:

1. R.B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi.
2. Dr. Kirpal Singh, Automobile Engineering (Vol. – 1), Standard Publishers Distributors, New Delhi.
3. S. Srinivasan, Automotive Mechanics, Tata McGraw Hill Education, New Delhi.
4. Automotive Mechanics N.K. Giri vol-2 Khanna Publishers, New Delhi
5. S.K. Gupta, Automobile Engineering, S.Chand, New Delhi.

Reference E-links

1. Clutch - https://www.youtube.com/watch?v=devo3kdSPQY
2. Manual Transmission - https://www.youtube.com/watch?v=wCu9W9xNwtI&t=29s
3. Automatic Transmission - https://www.youtube.com/watch?v=u_y1S8C0Hmc
4. Torque converter - https://www.youtube.com/watch?v=bRcDvCj_JPs
5. Differential - https://www.youtube.com/watch?v=nC6fsNXdcMQ

Proposed Syllabus of Fluid Mechanics & Machines

Course Code	AEPC402
Course Title	Fluid Mechanics & Machines
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	Applied Physics-I

Course Objectives:

1. To understand fluid and its properties.
2. To understand how to measure pressure, pressure difference, velocity and discharge in flow domains.
3. To understand Bernoulli's equation and its applications.
4. To understand head losses and its causes.
5. To understand working principle of centrifugal and reciprocating pumps.
6. To understand different aerodynamic effects on a vehicle.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1.	Fluid Properties and Measurement of Pressure: 1.1. Definition of fluid – from perspective of reaction to shear stress and lower cohesive forces. Fluid – liquid and gas. 1.2. Recap of fluid properties – Density, Specific gravity, Specific Weight, Specific Volume, Viscosity – dynamic and kinematic, Surface tension, Capillarity, Vapour Pressure, Compressibility (Definition, unit and application) 1.3. Ideal & real fluid. 1.4. Effect of temperature on Viscosity of both liquid and gas. (No explanation) 1.5. Newton's law of viscosity (Only formula and explanation, no numerical) 1.6. Capillary rise – derivation and simple numerical. 1.7. Pressure – Concept, unit, atmospheric pressure, absolute pressure, gauge pressure, vacuum pressure. 1.8. Pascal's law, derivation to show that intensity of pressure is same at a point from all direction, derivation of pressure-density-height relationship. Concept of head. Workout absolute value of atmospheric pressure. 1.9. Measurement of Pressure – Simple U-tube and differential manometer, Mechanical pressure gauge – Bourdon tube. 1.10. Simple numerical on application of simple U-tube and differential manometers where density of manometric fluid is more than working fluid.	**08**
2.	**Fluid Flow – Types, Measurements and Losses:** 2.1. Types of fluid flows - Steady & unsteady, laminar& turbulent. Reynold's number – explanation and expression. Concept of streamline. Reynold's number ranges for laminar and turbulent flows in circular pipes. 2.2. Continuity equation: - Conservation of mass, derivation, assumptions, expressions in case of compressible and incompressible fluids, discharge.	**15**

	Simple numerical with bifurcation of pipe. 2.3. Bernoulli's equation: - Conservation of energy, derivation, assumptions, application to real fluids, head loss. 2.4. Discharge – expression, unit. 2.5. Hydraulic coefficients –vena-contracta, coefficients of discharge, velocity and contraction. Relation among those. 2.6. Measurement of discharge –Venturimeter and Orificemeter, construction, derivation of discharge in venturimeter only, nozzle and diffuser effects, flow separation and formation of eddies in diverging section of venturimeter. Divergence angle. Simple numerical involving both the meters. 2.7. Measurement of velocity – Simple Pitot tube, stagnation pressure, static pressure. Simple numerical. 2.8. Head loss in pipes – Major & minor loss, sources, formula – Darcy-Weisbach (no derivation), friction coefficient, friction factor.	
3.	**Fluid Machines – Pumps:** 3.1. Pump – definition, classification – dynamic, positive displacement, rotary, reciprocating etc. 3.2. Centrifugal Pump – Construction, working principle, application, priming, types of impeller & casing, diffuser vanes. Static head, manometric head, work done, different efficiencies. Cavitation – concept &measures to prevent cavitation. Simple numerical on power required and overall efficiency. 3.3. Staging – parallel & series (concept only) 3.4. Reciprocating Pump – Construction, single acting, double acting, working principle, application; derivation of discharge & power required; slip – positive and negative, reason of negative slip, percentage of slip; Simple numerical on discharge, power required, slip etc.	**09**
4.	**Introduction to Vehicle Aerodynamics:** 4.1. Aerodynamics –Need of Vehicle Aerodynamics, objectives of vehicle aerodynamics, road resistance (total road loads), air resistance, rolling resistance. 4.2. Aerodynamic forces-Drag, types of drag, drag coefficient – effect of drag, aerodynamic aids - drag side force, lift force, effect of lift force- pitching, yawing, rolling moment 4.3. Crosswind sensitivity - recirculating flow (vortex) – diffusers - vortex noise 4.4. Wind shield – Types, application	**06**
5.	**Aerodynamic Effects on Vehicle Body:** 5.1. Aerofoil- vehicle as a bluff body - Mechanics of air flow - pressure distribution on a vehicle 5.2. Methods Used for Evaluating Vehicle Aerodynamics (Name, Definition) 5.3. Car Classifications-Different types of car body shape (name, Example)- Automobile Shapes and Aerodynamics - 5.4. Passenger Cars - Convertible (Cabriolet) and Semi-Convertible (Cabrio Coach), Coupe, Crossover (CUV), Hatchback, Landaulet (Landaulette), Limousine (Limo), Roadster, Sedan, Sport Utility Vehicle (SUV), Wagon (Station Wagon) (Definition, Example)	**04**

Total Hours	42

Weightage distribution in both objective, short and broad answer type questions:

Group	Module Number	Weightage (%)
A	1 & 2	50
B	3	20
C	4 & 5	30

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Compute capillary rise, compute pressure at a point and pressure difference between two points and compute discharge in case of bifurcation of pipes
CO2	Compute coefficient of discharge, discharge using venturimeter/orificemeter and calculate velocity using pitot tube and estimate losses in pipes.
CO3	Calculate discharge, slip and power required in reciprocating pump and power required and overall efficiency in centrifugal pumps. Identify cause of cavitation.
CO4	Identify the different aerodynamics forces on a vehicle.
CO5	Classify different shapes of automobile body.

Text Books:

1. Fluid Mechanics and Fluid Power Engineering – Dr, D. S Kumar, Kataria and Sons
2. Fluid Mechanics and Hydraulic Machines – R. K. Bansal, Laxmi Publications, New Delhi
3. Fluid Mechanics & Hydraulic Machines, S.S. Rattan, Khanna Publishing House, New Delhi
4. Hydraulic, fluid mechanics & fluid machines – Ramamrutham S, Dhanpath Rai and Sons, New Delhi.

Text Book for Unit 4 & 5:

Aerodynamics of Road vehicles, W. H. Hucho, Butterworths Co. Ltd., 1997

Reference Books:

1. Automotive Aerodynamics: Update SP-706, SAE, 1987.
2. Vehicle Aerodynamics, SP-1145, SAE, 1996.
3. Fluid Mechanics – Munson, Okiishi *et al.* –Wiely
4. Fluid Mechanics in SI units – F. M. White – McGraw Hill

Proposed Syllabus of Automotive Component Design

Course Code	AEPC403
Course Title	Automotive Component Design
Number of Credits and L-T-P	3 [L – 2, T – 1, P - 0]
Course Category	PC
Prerequisites	Engineering Mechanics, Strength of Materials

Course objectives:

After completing this course, the students will be able

1. To enable the student to design and draw simple machine components used in Automobile.
2. To understand the basic philosophy and fundamentals of Machine Design.
3. To understand the modes of failures of m/c components and decide the design criteria and equations.
4. To analyze and evaluate the loads, forces, stresses involved in components and subassemblies and decide the dimensions.
5. To develop analytical abilities to give solutions to engineering design problems.

Module No.	Description of Topic	Contact Hrs.
1	**Introduction to Design** 1.1. Introduction to design, Classification of design, Design consideration & Design procedure. 1.2. Stress analysis - Types of external loads – 1.3. Types of induced stresses - tensile – compressive – shear - Crushing and bearing pressure – bending – torsion - thermal stresses – creep- proof stresses- resilience -principal stresses. 1.4. Variable stresses in machine parts, fatigue & endurance limit, stress– Time diagrams for variable stresses - Working stresses for static load - variable or fatigue load - Stress concentration causes and remedies. 1.5. Designation of materials as per IS and introduction to International standards & advantages of standardization, use of standards in design and preferred numbers series. 1.6. Selection of material and justifications for Automobile components - Advanced Materials for automotive components. 1.7. Post design aspects - Ergonomic aspect, Aesthetic consideration (Shape, colour, surface finish) for Automobile	**08**
2	**Design of Shafts, Keys and Levers** 2.1. Conceptual understanding of shaft, axles & spindles. 2.2. Design of shaft for torsion, rigidity, bending, combined torsion & bending. [simple numerical] 2.3. Design of propeller shaft, whirling & critical speed.[No Numerical] 2.4. Rear axle – design consideration –martial [No Numerical] 2.5. Types of keys, design of Sunk Rectangular Key, Effect of keyways on strength of shaft. [simple numerical] 2.6. Types of levers – application of levers in automobile.	**08**

3	**Design of Joint** 3.1. Cotter joint- Application- Advantages- Terminology - Design consideration and Design process of Cotter joint. [simple Numerical] 3.2. Knuckle joint- Application – advantages- Terminology -Design consideration. [No Numerical]	**06**
4	**Bearing and Springs** 4.1. Bearings - location in Automobiles systems 4.2. Classification of Bearings - Sliding contact - Rolling contact – Terminology. 4.3. Life Load relationship - Basic static load rating and Basic dynamic load rating - limiting speed of ball bearing. [No Numerical] 4.4. Spring – Type of Spring – Application – Helical spring - Multi leaf spring – Terminology of spring – Spring Materials [definition only]	**05**
5	**Design of Couplings** 5.1. Coupling- Oldham coupling-Hooke's coupling (definition) 5.2. Muff Coupling- Advantages- Disadvantages – Terminology. 5.3. Clamp Coupling – Advantages – Disadvantages – Terminology. 5.4. Flange Coupling (Rigid)-Advantages – Disadvantages – Terminology - Design Procedure for Flange Coupling. [Simple Numerical]	**06**
6	**Design of IC Engine Parts** 6.1. Analyze the engine specifications and calculations of cylinder dimensions for given power. – cylinder liner- cylinder bore_ cylinder length 6.2. Design of cylinder head thickness and bolts- cylinder wall thickness. 6.3. Piston- piston material-Design of piston crown by bending strength and thermal considerations. 6.4. Design of piston rings and skirt length. 6.5. Design of piston pin for bearing - bending & shear considerations. 6.6. Simple numerical on cylinder – cylinder head – piston	**09**
Total Hours		**42**

Weightage distribution in both objective, short and broad answer type questions:

Group Name	Module Number	Weightage (%)
A	1 & 2	40
B	3 & 4	20
C	5 & 6	40

Course Outcome:

At the end of the course, the student will be able to:

CO1	Distinguish different type of force act in a loaded system.
CO2	Design shaft, key and engine component in different load condition.
CO3	Identify the different Automobile component and their application.
CO4	Able to grasp the concept of designing of different Automobile part.
CO5	Analyze various forces coming on the piston and design the process of the component.

Text Books:

1. Introduction to Machine Design – V. B. Bhandari, Tata Mc- Graw Hill, New Delhi.
2. Machine design – R. K. Jain, Khanna Publication, New Delhi.
3. Machine design – Pandya & Shah, Dhanpat Rai & Son, New Delhi.
4. A Text Book of Machine Design, Khurmi R S, Gupta J K, Eurasia Publishing house, New Delhi, 2010.

Reference Books:

1. Machine Design – Sadhu Singh, Khanna Book Publishing Co., Delhi (ISBN: 978-9382609-575)
2. Machine Design Data Book – Sadhu Singh, Revised Edition, Khanna Book Publishing Co., Delhi (ISBN: 978-9382609-513).
3. Mechanical Engineering Design – Joseph Edward Shigley, Tata Mc- Graw Hill, New Delhi.
4. Machine design – Pandya & Shah, Dhanpat Rai & Son, New Delhi.
5. Design Data Book – PSG Coimbtore, PSG Coimbtore.
6. Hand Book of Properties of Engineering Materials & Design Data for Machine Elements – Abdulla Shariff, Dhanpat Rai & Sons, New Delhi.

Proposed Syllabus of Theory of Machines & Mechanisms

Course Code	AEPC404
Course Title	Theory of Machines & Mechanisms
Credits and L-T-P	3 [L – 2, T – 1, P - 0]
Course Category	PC
Prerequisites	Engineering Mechanics

Course Objectives:

1. To impart basic knowledge of theory of machine
2. To understand construction of cam profiles for desired motion of followers
3. To understand methods of power transmission and field of applications
4. To understand construction and working of flywheel and governors
5. To understand construction and working of brakes & dynamometers.
6. To understand functions of clutches and bearings, related theories.
7. To understand the importance and methods of balancing, causes and harmful effects of vibrations.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1.	Basic Kinematics: 1.1 Introduction to kinematics, Machine & structure. 1.2 Rigid & Resistant body; 1.3 Kinematic link - its types, kinematic pair-its types, Kinematic chain, types of joints in a chain; 1.4 Mechanism, inversion of mechanism, four bar chain- single slider crank mechanism only. (No expression/ derivation, no numerical)	**03**
2.	**Cams and Followers:** 2.1 Concept; Definition and application of Cams and Followers; 2.2 Classification of Cams and Followers; 2.3 Follower motions and their displacement diagrams like uniform velocity, SHM, uniform acceleration and Retardation (no discussion on expressions) 2.4 Drawing of profile of radial cam with knife-edge and roller follower with and without offset with reciprocating motion (graphical method only, no numerical for calculation of velocity or acceleration/retardation).	**05**
3.	**Power Transmission:** 3.1 Types of Drives – Belt, Chain, Rope, Gear drives & their comparison; 3.2 Belt Drives - Flat belt, V– belt & its applications; Material for flat and V-belt; (Simple numerical); 3.3 Angle of lap, Belt length, Slip and Creep, Law of Belting, Determination of Velocity Ratio, Ratio of tight side and slack side tension; Centrifugal tension and Initial tension; Condition for maximum power transmission (Simple numerical); 3.4Gear Drives – Spur gear terminology; Types of gears and gear trains, their selection for different applications; Train value & Velocity ratio for simple, compound, reverted and simple epicyclic gear trains; Calculation of gear ratio,	**12**

	centre distance between two shafts, Torque & power transmission; Methods of lubrication;[simple numerical] 3.5 Chain Drives –Selection of Chain & Sprocket wheels; Methods of lubrication; Advantages & Disadvantages; [no numerical] 3.6 Rope Drives – Types, applications, advantages & limitations of Steel ropes.[no numerical]	
4.	**Flywheel and Governors:** 4.1 Flywheel - Concept, function and application of flywheel with help of turning moment diagram for single cylinder 4-Stroke I.C. Engine (simple numerical); 4.2 Coefficient of fluctuation of energy, Coefficient of fluctuation of speed and its significance; Determination of mass of flywheel and energy stored in a flywheel.(simple numerical) 4.3Governors- Types and explanation with neat sketches (Centrifugal, Watt and Porter); Concept, function and applications & Terminology of Governors; (No deduction on related expressions and no numerical) 4.3 Comparison between Flywheel and Governor.	**08**
5.	**Brakes & Dynamometers:** 5.1 Function of brakes and dynamometers; 5.2 Types of brakes and Dynamometers; Comparison between brakes and dynamometers; 5.3 Construction and working of i) shoe brake, ii) Band Brake, iii) Internal expanding shoe brake iv) Disc Brake; Concept of Self Locking & Self energizing brakes; 5.4 Numerical problems to find braking force and braking torque for shoe & band brakes; 5.5 Construction and working of i) Rope Brake Dynamometer,[no deduction only formula based numerical problem] ii) Hydraulic Dynamometer,[no numerical] iii) Eddy current Dynamometers; ,[no numerical]	**06**
6.	**Clutches & Bearings:** 6.1 Clutches- Uniform pressure and Uniform Wear theories; Function of Clutch and its application; (no derivation and simple numerical using formula) 6.2 Construction and working of i) Single plate clutch, ii) Multi-plate clutch, iii) Centrifugal Clutch iv) Cone clutch and v) Diaphragm clutch. 6.3 Bearings – i) Simple Pivot, ii) Collar Bearing, iii) Conical pivot. (descriptions only)	**04**
7.	**Balancing & Vibrations:** 7.1 Concept of balancing; 7.2 Balancing of single rotating mass; analytical method for balancing of several masses revolving in same plane; [simple numerical] 7.3 Concept and terminology used in vibrations, Causes of vibrations in machines; their harmful effects and remedies.	**04**
Total Hours		**42**

Weightage distribution in both objective, short and broad answer type questions:

Group	Module Number	Weightage (%)
A	1 & 2	25
B	3 & 4	45
C	5, 6, 7	30

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Describe working principle, classification, if any & applications of kinematic links, kinematic chains, joints& slider crank mechanism.
CO2	Explain different types of cam & follower along with to draw cam profiles for desired follower motion
CO3	Explain different type of belt and gear drive along with to solve simple numerical problems
CO4	Describe working of flywheel and governors along with calculation of flywheel mass and energy stored.
CO5	Explain the working of different types of brake and dynamometer and to be able to calculate braking force and torque.
CO6	Explain the constructional features & working principle of clutches and bearings.
CO7	Describe & calculate balancing mass required to reduce vibration & noise, along with the harmful effects of vibration and remedies.

Text Books:

1. Theory of Machines, Khurmi Gupta, Eurasia publishing House Pvt. Ltd.
2. Theory of Machines, S.S. Rattan, McGraw Hill companies

References Books:

1. Theory of Machines, P.L. Ballaney, Khanna Publication
2. Theory of Machines, Jagdishlal, Bombay metro-politan book limited
3. Theory of Machines, Ghosh – Mallik, Affiliated East west press
4. Theory of Machines, Thomas Bevan, Pearson
5. Theory of Machines, J.E. Shigley, Oxford
6. Theory of Machines, V.P.Singh, Dhanpat Rai & Co.

Proposed Syllabus of Automotive Pollution & Control

Course Code	AEPE411
Course Title	Automotive Pollution & Control
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PE
Prerequisites	Environmental Science and Engineering, Basic theory of IC Engine

Course Objectives:

1. To develop the basic knowledge of the students in automobile engines pollution formation & control techniques, Measurement techniques.
2. Know the social, cultural, global and environmental responsibilities of the professional engineer, and the principles of sustainable design and development.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1.	**Introduction** Vehicle population assessment in metropolitan cities and contribution to pollution, effects on human health and environment, global warming, types of emission, transient operational effects on pollution.	**04**
2.	**Pollutant Formation in SI Engines** Pollutant formation in SI Engines, mechanism of HC and CO formation in four stroke and two stroke SI engines, NO_x formation in SI engines, effects of design and operating variables on emission formation. Two stroke engine pollution.	**06**
3.	**Pollutant Formation in CI Engines** Pollutant formation in CI engines, smoke and particulate emissions in CI engines, effects of design and operating variables on CI engine emissions. NO_x formation. Noise pollution from automobiles	**06**
4.	**Control of Emissions from SI and CI Engines** Controlling of pollutant formation in engines exhaust after treatment, charcoal canister control for evaporative emission control, emissions and drivability, positive crank case ventilation system for UBHC emission reduction, exhaust gas recirculation, exhaust after treatment – DOC, DPF, SCR and LNT measurement.	**10**
5.	**Measurement Techniques, Test Procedures & Emission Standards** NDIR analyzers, FID, chemiluminescence NO_x analyzer, oxygen analyzer, smoke measurement, constant volume sampling, particulate emission measurement, orsat apparatus. (Numerical based on formula of orsat apparatus). Test Cycles for Light, Medium & Heavy duty vehicle engines, USEPA, European, Japan & India driving cycles. Emission Standards for Light and Heavy duty engines. Emission Norms: As per	**12**

	Bharat Standard up to BS – VI.	
6.	**Health Effects of Emissions from Automobiles** Emission effects on health and environment. Emission inventory, ambient air quality monitoring.	**04**
Total Hours		**42**

Weightage distribution in both objective, short and long answer type questions:

Group Name	Unit Number	Weightage (%)
A	1 & 2	30
B	3 & 4	40
C	5 & 6	30

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Describe the emission and its effect on human health and environment.
CO2	Identify the formation of pollutant in SI engine.
CO3	Identify the formation of pollutant in CI engine
CO4	Describe the Emission control techniques.
CO5	Describe the Emission measurement techniques, Emission Standards and various test procedure

Text Books:

1. Engine Emission, B.P Pundir, Narosa Publication.
2. Automotive Pollution & Control, A. K. Babu, AKB Publishers

Reference Books:

1. Internal Combustion Engines, V .Ganesan, Tata McGraw Hill.
2. Automotive Emission Control, Crouse, W.M. and. Anglin, A.L, McGraw Hill.
3. IC Engines, Dr. S. S. Thipse, Jaico publications.
4. Engine Emissions, pollutant formation, G.S .Springer and D.J. Patterson, Plenum Press.
5. ARAI vehicle emission test manual.

Proposed Syllabus of Alternate Fuels and Energy Systems

Course Code	AEPE412
Course Title	Alternate Fuels and Energy Systems
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PE
Prerequisites	Engineering Chemistry, Basic theory of IC engine

Course Objectives:

1. Differentiate the conventional fuels and Alternative fuels.
2. Understand the future fuel and energy systems for automobile applications in the context of production, storage, efficiency and emission.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Introduction to Alternative Fuel** Introduction to alternative fuels, Need for alternative fuels, Availability of different alternative fuels for SI and CI engines, Selection criterion of fuel, Selection criteria of raw material.	**04**
2	**Alcohol** Introduction, Raw Material Selection criteria for Methanol and Ethanol, Potential of Methanol and Ethanol (Petrol Blends and Diesel Blends, Bio diesel production ingredient, Di methyl ether, Fuel cell) Production method of ethanol, Production methods of methanol, Methanol Economics, Methanol and Ethanol Safety Aspects, Properties of Methanol and Ethanol, Methanol and Ethanol Engine tests, Methanol and Ethanol Benefits, Engine Modification required for alcohol uses in SI and CI engine.	**08**
3	**Biodiesel** Introduction, Raw materials used for production of Bio Diesel, Biodiesel production methods, Bio Diesel Production Equipment, Types of Transesterification process, Bio Diesel Properties, Bio Diesel Quality Standards, Performance characteristics of biodiesel, Engine Tests for Bio Diesels, Engine Modification required for biodiesel uses in engine, Challenges for Bio Diesel.	**07**
4	**Biogas** Introduction, Biogas production method, Biogas Plants, Factors affecting biogas formation, Engine Modification required for biogas uses in SI and CI engine, Biogas purification methods, Emission Characteristics of biogas, Storage and safety aspect of Biogas.	**07**
5	**CNG & LPG and Hydrogen** Potential of LPG, LPG production, Properties of LPG, LPG Modelling studies, LPG Engine tests, LPG Material compatibility, LPG Economics, LPG Safety aspects, LPG Merits and Demerits, CNG Storage, CNG Distribution, CNG Safety, CNG Advantages, CNG challenges, Hydrogen Production, Hydrogen Storage,	**08**

	Hydrogen Properties, Hydrogen Economics, Hydrogen Safety, Hydrogen in Fuel cell, Hydrogen in SI and CI engines, Hydrogen Benefits, Hydrogen Barriers and challenges	
6	**Electric, Hybrid, Fuel cell and Solar Vehicle** Introduction, Need of Electric, Hybrid, Fuel cell and Solar vehicle, Principle of Electric Vehicles, Construction of Electric vehicles, Benefits and Challenges of Electric Vehicle, Fuel Cell Vehicles, Types of Fuel cell vehicle, Topological configuration of electric, hybrid vehicle.	**08**
Total Hours		**42**

Weightage distribution in both objective, short and long answer type questions:

Group Name	Unit Number	Weightage (%)
A	1 & 2	35
B	3 & 4	35
C	5 & 6	30

Course Outcomes:

At the end of the course, the student will be able to:

CO1	To understand broad comprehension of future alternative transportation fuels: alcohol, biodiesel, biogas and their production technologies.
CO2	To understand environmental assessment and performance assessment of alternative fuels.
CO3	To understand Electric, Hybrid fuel cell and solar vehicle technologies and their economic consideration
CO4	To apply various methods for alternative fuel usages and engine modification requirement in automobile engines.

Text Books:

1. Alternative Fuels, Dr. S. S. Thipse, Jaico publications.
2. Electric & Hybrid Vehicles, A.K. Babu, Khanna Publishing.

Reference Books:

1. Alternative Fuels for Transportation, A. S. Ramadhas, CRC Press.
2. Internal Combustion Engines, V .Ganesan, Tata McGraw Hill.
3. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design, Mehrdad Ehsani, Yimin Gao, Ali Emadi, CRC Press.
4. Electric and Hybrid Vehicles, Tom Denton, Routledge.

Proposed Syllabus of Automotive Powertrain Laboratory

Course Code	AEPC405
Course Title	Automotive Powertrain Laboratory
Number of Credits and L-T-P	1 [L – 0, T – 0, P - 2]
Course Category	PC
Prerequisites	NIL

Course Objectives:

1. To prepare Vehicle Layout and Automotive Powertrain System
2. To find out the process of Clutch engaging and disengaging mechanism.
3. To identify different gear ratio of Manual transmission system.
4. To know about Automatic Transmission System
5. To distinguish between different type of joints
6. To know about the Differential

Course Contents:

Exp. No.	Description of Experiments	Contact hours
1.	Understand the layout, of "Vehicle Layout and Automotive Powertrain System laboratory", make use of various tools, and measuring devices, write their specifications, application and care to be taken while using the same.	**04**
2.	Dismantle and assemble a single plate dry type clutch assembly, to understand it's construction and working. Sketch and label all the components.	**04**
3.	Dismantle and assemble Torque Converter assembly, to understand construction and working. sketch and label all of its components.	**04**
4.	Dismantle and assemble a Synchromesh gearbox, to understand its construction and working. Observe gear shifting (synchronizing action), draw power flow diagrams, and calculate gear ratios of different gears (1st, 2nd, 3rd , 4th. 5th Gear & Reverse).	**06**
5.	Observe and draw Automatic transmission layout of a Four wheeler vehicle	**04**
6.	Dismantle and assemble a Propeller shaft, Slip joint, CV joint and Universal Joint, to understand their construction and working. Sketch the same.	**06**
7.	Dismantle and assemble the open and Limited Slip Differential, to understand its construction and working. Sketch the unit showing the exact location of the bearings. Also find the gear ratio of final drive	**04**
8.	Dismantle and assemble the front and Rear axle, to understand its construction and working and also find out the type of axle	**04**
Total Hours		**28**

Experiment Number 1, 2, 4 & 6 are compulsory and any two from the serial number 3, 5, 7 & 8.

Course Outcomes:

At the end of the course, students will be able to:

CO1	Identify basic tools of Powertrain Laboratory
CO2	Dissemble and assemble vehicle transmission system

CO3	Identify different types of gears and gear ratios
CO4	Able to grasp the construction and working procedure of Manual & Automatic transmission system
CO5	Identify the construction and working of Differential

Text Books:

1. R.B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi.
2. Dr. Kirpal Singh, Automobile Engineering (Vol. – 1), Standard Publishers Distributors, New Delhi.
3. S. Srinivasan, Automotive Mechanics, Tata McGraw Hill Education, New Delhi.
4. Automotive Mechanics N.K. Giri vol-2 Khanna Publishers, New Delhi
5. S.K. Gupta, Automobile Engineering, S.Chand, New Delhi.

Proposed Syllabus of Strength of Materials & Fluid Mechanics Laboratory

Course Code	AEPC406
Course Title	Strength of Materials & Fluid Mechanics Laboratory
Number of Credits and L-T-P	1.5 [L – 0, T – 0, P - 3]
Course Category	PC
Prerequisites	Strength of Materials, Fluid Mechanics & Machines

Course Objectives:

1. To experimentally find out hardness and toughness of metallic objects.
2. To find out various points of interest from stress-strain diagram of a mild steel specimen.
3. To experimentally find out shear modulus of metallic objects.
4. To experimentally verify Bernoulli's equation.
5. To experimentally find out discharge coefficient of venturimeter/ orificemeter.
6. To identify main components of centrifugal pump.

Course Contents:

Module No.	Description of Experiments	To be Done	Number of hours
1.	**Experiments from Strength of Materials:** 1.1. Experimental determination of hardness of a metallic specimen using either Rockwell/ Brinell hardness test. 1.2. Experimental determination of impact strength (toughness) of a metallic specimen using either Charpy/ Izod impact test. 1.3. Tensile test of a mild steel specimen using UTM and draw corresponding stress-train diagram and identify yield stress, ultimate stress, breaking stress, percentage elongation in length (measure of ductility) and determination of modulus of elasticity. 1.4. Conduct torsion test on a mild steel bar and find the value of shear modulus G. 1.5. Find strength and stiffness of a spring.	**At least three**	**16**
2.	**Experiments from Fluid Mechanics & Machinery:** 2.1. Study of Bourdon tube pressure gauge. 2.2. Experimental determination of coefficient of discharge of venturimeter/ orificemeter. 2.3. Experimental verification of Bernoulli's theorem. 2.4. Dismantling and assembly of a centrifugal pump and identify its main components. 2.5. Trial on a centrifugal pump to determine efficiency.	**At least three**	**26**
Total Hours		**42**	

Course Outcomes:

At the end of the course, students will be able to:

CO1	Calculate hardness, toughness and shear modulus of metallic samples.
CO2	Calculate elastic modulus, ultimate strength and measure ductility of metallic samples.

CO3	Calculate coefficient of discharge of venturi/orifice meter.
CO4	Identify different components of a centrifugal pump.
CO5	Verify experimentally the Bernoulli's equation.

Text Books:

1. Fluid Mechanics and Machinery Laboratory Manual, N. Kumara Swamy, Charotar Publishing House Pvt. Ltd., ANAND 388 001, Ed. 2008
2. Strength of Materials – S.S Bhavikatti, Vikas Publishers
3. Strength of Materials – U.C. Jindal, Pearson
4. Strength of Materials – R.S. Khurmi, S.Chand Company Ltd. Delhi
5. A Text Book Strength of Material– R.K. Bansal, Laxmi Publication New Delhi

Proposed Syllabus of Automobile Engineering Drawing and Computer Graphics Laboratory

Course Code	AEPC407
Course Title	Automobile Engineering Drawing and Computer Graphics Laboratory
Number of Credits and L-T-P	2 [L – 0, T – 0, P - 4]
Course Category	PC
Prerequisites	Engineering Drawing, Strength of Material, Automotive Component Design

Course Objectives:

1. To prepare drawing sheet with proper dimension and title box on CAD
2. To find out the process of design on CAD.
3. To identify different commend and tools and there use on CAD.
4. To know about designing process of flange coupling.
5. To distinguish between CAD drawing and hand drawing.

Course Contents:

Exp. No.	Description of Experiments	To be Done	Number of hours
1.	Introduction of CAD and brief description about basic commend, tools, absolute coordinate system and relative coordinate system.	**Compulsory**	**6**
2.	Line, poly line, rectangle, arc line, different type of line, circle, Move ,Rotate, Scale, copy, Mirror, erase, trim, extend, Annotate Dimension Style on CAD		**4**
3.	Single line text, Multiline text, Layer properties, Insert blocks, Dimensional Manage, Drawing units, Sheet settings on CAD		**6**
4.	Isometric views - Isometric top, left, right and Isometric diagrams on CAD.		**4**
5.	Calculate the dimension of a shaft for given problem (problem given by the faculty) and design it in a drawing sheet and on CAD.	**At least three**	**12**
6.	Calculate the dimension of Flange Coupling for given problem (problem given by the faculty) and design it in a drawing sheet and on CAD.		**12**
7.	Calculate the dimension of Flange Cotter joint for given problem (problem given by the faculty) and design it in a drawing sheet and on CAD.		**12**
8.	Calculate the dimension of Engine block (cylinder, piston) joint for given problem (problem given by the faculty) and design it in a drawing sheet and on CAD.		**12**
Total Hours		**56**	

Course Outcomes:

At the end of the course, students will be able to:

CO1	Identify basic tools of CAD
CO2	Draw isometric view of an object on CAD

CO3	Arrange for proper dimension and text in a CAD drawing.
CO4	Draw the dimension of a joint (Cotter/ Flange Coupling).
CO5	Identify different tolerances and consideration for design

Text Books:

1. Engineering Drawing with Auto CAD – B. V. R Gupta & M. Raja Roy, I K International Publication House
2. AutoCAD 2019 Training Guide - Linkan Sagar & Nisha Gupta, BPB; First, 2019 edition
3. A Hand Book on AutoCAD Tools Practice - Azhar Wahab, Notion Press; 1st edition
4. Learn AutoCAD in a Easy Way, Sunil K. Pandey, S.K. Kataria& Sons; Reprint 2010 edition
5. Engineering Drawing + Autocad, K. Venugopal, New Age; Fifth edition
6. Introduction to Machine Design – V. B. Bhandari, Tata Mc- Graw Hill, New Delhi.
7. Machine design – R. K. Jain, Khanna Publication, New Delhi.
8. Auto design problems, K M Aggarwal, Satyaprakashan

Proposed Syllabus of Minor Project

Course Code	PR202
Course Title	Minor Project
Number of Credits and L-T-P	2 [L – 0, T – 0, P - 4]
Course Category	Minor Project
Prerequisites	All subject up-to 4th Semester

Course objectives:

After completing this course, the students will be able

1. To understand the solution process of real life problem
2. To achieve the potentiality of doing team work
3. To understand the gap between academic knowledge and actual real life problem solving knowledge.
4. To prepare the project repot in a skill full way.

Project group:

- Formation of project group: Maximum 8 to 12 students per batch.
- Each project group should select work by consulting the guide.

Course Content:

During this minor project, two projects have to be completed by each group.
One project is on an advanced topic of Automotive Engine or Automotive Transmission

Some suggested topics are:

- Design considerations of Combustion Chamber in CI engine and parameters to control abnormal combustion.
- Solar powered vehicle
- Conversion of petrol vehicle into LPG vehicle by using LPG kit
- Electromagnetic clutch
- Dual Clutch Transmission (DCT)
- Preparation of Bio-Diesel.
- Ethanol Blended Petrol and its effect on SI engine
- Torque Vectoring Differential
- Automatic tyre pressure inflation system
- Smart antitheft system for vehicle security

and

One project is about any subject within the 1st to 4th semester syllabus.

Some Suggested Topics are:

- Forming Aerodynamics profile for any type of passenger car.
- Propeller shaft designing for front wheel, rear wheel and all-wheel drive in CAD.

- Advanced surface coating techniques like chemical vapor deposition, ion implantation, and physical vapor deposition.
- Engine Performance and testing
- Hybrid Car
- Electric car
- Fuel cell Electric Vehicle (FCEV)
- Electronic Stability Program (ESP)
- Different Vehicle testing processes
- Maintenance of Automobile air-conditioning systems.

Faculty Members can select any project topic from out of the suggested topic but within the syllabus.

After completion of the project, each students group should prepare two (one for each project) comprehensive report to indicate what are observed and learnt during the project work. The student may contact guide for assigning topics and problems and should prepare the final report on the assigned topics. The project report should be signed by the guide and HOD.

The format of the project report will be as following

Sl. No.	Particulates
1	Title page
2	Deceleration page
3	Acknowledgement
4	Certificate from guide
5	Abstract
6	Objective
7	Literature review/ background survey/history
8	Present work
9	Methodology
10	Observation
11	Conclusion
12	References

Internal Assessment:

1. Project Report: The project Report will be evaluated on the basis of following criteria (as applicable) 40 Marks

Sl. No.	Criteria for evaluation of Internship Report
1	Originality
2	Adequacy and purposeful write up
3	Organizations, format, drawing, sketches, style, language
4	Practical applications and relationships with basic theory
5	Concept taught in the outcome

2. Attendance and work process 20 Marks

External Assessment:

Seminars must be arranged for the student based on the project report, in presence of project guide, Internal Committee constituted by the concerned department of the Institute and External examiner/s. The evaluation will be based on the following criteria. 40 Marks

Sl. No.	Criteria for evaluation of Internship Report
1	Quality of content presented
2	Proper Planning for presentation
3	Effectiveness of presentation
4	Depth of knowledge and skills
5	Viva voce

Course Outcome:

At the end of the course, the student will be able to:

CO1	learn new skills and supplement knowledge
CO2	Learn & practice communication and teamwork skills.
CO3	Learn strategies like time management, multi-tasking, real time technical knowledge etc.
CO4	Can apply their knowledge for doing some application oriented work.
CO5	Gain practical experience in a real working environment.

West Bengal State Council of Technical &Vocational Education and SkillDevelopment (TechnicalEducationDivision)

Syllabus
of

DiplomainAutomobileEngineering[AE]

Part-III(5th Semester)

Revised2022

Semester-wiseDetailedCurriculum

Semester V(Thirdyear)Branch/Course:Automobile Engineering

Sl. No.	Category	Code	CourseTitle	Hoursperweek			Totalcontacthours/ week	Credits
				Lecture	Tutorial	Practical		
1	Program CoreCourse	AEPC 501	FluidPower	3	0	0	3	3
2	ProgramCoreCourse	AEPC502	Automotive Electrical& Electronics	3	0	0	3	3
3	ProgramCoreCourse	AEPC503	M V Act &Transport Management	3	0	0	3	3
4	Program ElectiveCourse	AEPE 511/ 512	Anyone ProgrammeElective	3	0	0	3	3
5	Program ElectiveCourse	AEPE 513/ 514	Anyone ProgrammeElective	3	0	0	3	3
6	ProgramCoreCourse	AEPC504	Automotive Electrical &ElectronicsLab	0	0	2	2	1
7	Internship-IIafterIVth Sem	SI501		0	0	0	0	1
8	Major Project	PR502		0	0	2	2	1
Total				**15**	**0**	**4**	**19**	**18**

ListofProgramElectivesforFifthSemester

1. AEPE511:ModernManufacturingMethods
2. AEPE512:AutomobileAirConditioning
3. AEPE513:EarthMovingEquipments&FarmMachinery
4. AEPE514:Two&ThreeWheelers

Proposed Syllabus of Fluid Power

CourseCode	AEPC501
CourseTitle	FluidPower
NumberofCreditsandL-T-P	3[L–3,T–0,P-0]
CourseCategory	PC
Prerequisites	AppliedPhysics-I

CourseObjectives:

1. TounderstandworkingprincipleofFluidPowerSystemsanditsbasiccomponents
2. Tounderstandworkingprincipleofgear,vane&pistonpumps&theiradvantages&disad vantages
3. TounderstandworkingprincipleofvariouscontrolcomponentsofFluidPowersystemsnam elydirection,flowandpressurecontrolvalves.
4. Tounderstandworkingprincipleofbothlinear&rotaryactuators.
5. TounderstandbasiccomponentsofPneumaticsystems.
6. Tounderstandworkingprincipleofcommonhydraulic&pneumaticcircuits
7. ToidentifyapplicationareasofFluidPowerSystemsinautomobiles.

CourseContents:

Module No.	Description of Topic	Contact Hrs.
1.	**Introduction to Fluid Power:** FluidPower–Definition,Principle–Pascal'sLaw,Types–Hydraulics&Pneumatics,Applicationingeneral,ApplicationinAutomotive industry. Hydraulics – Definition, Basic components of a hydraulic system,Areasofapplication,advantages&disadvantages Pneumatics – Definition, Basic components of a pneumatic system,Areasofapplication,advantages&disadvantages ComparisonbetweenHydraulics&Pneumatics	**04**
2.	**Hydraulics–Pumps:** Positivedisplacementpumptypes–Gear,Vane&Pistonpumps External Gear Pump – Construction, working principle, Workingpressure range, Application, Advantages & disadvantages. Symbol.Expressionfortheoreticaldischarge,Simpleproblems. VanePump–Constructionofbothbalanced&unbalanced,workingprinciple,Workingpressurerange,Application,Advantages&disadvantages.Symbol. Piston Pump – Types – Axial (both swash plate & bent axis design)&Radial,Construction,workingprinciple,Workingpressurerange, Application,Advantages&disadvantages.Symbol.	**06**

3.	**Hydraulics-ControlComponents:** Direction Control Valve: Function, Check valve – ball type, poppettype, pilot operated. 3/2, 4/2, 4/3 DCV application; Different typesof neutral positions – closed, tandem, float, open; various types ofactuations– Manual,Solenoidetc.Symbols. FlowControlValve:Types– nonpressurecompensatedandpressurecompensated,Applicationofbothtypes.Symbols. PressureControlValve:Types– Relief,reducing,unloading&Sequencevalve.Application,Symbols. Servo&ProportionalControlValves–Electro-hydraulicservovalve and4-way proportional control valve, use,symbols and application areas.	**09**
4.	**HydraulicsActuators:** Actuators–Linear&rotary,Applications Linear Actuator – Cylinder, Single & Double acting. Construction ofdoubleactinghydrauliccylinder.Creepofcylinder.Cylindercushioning. RotaryActuator– Hydraulicmotors,Limitedrotationandfullrotation,gear,vaneandpistonmotors.Symbolsandapplication areas.	**04**
5.	**Hydraulics–Accessories:** 5.1. Reservoirs,Pipes,hoses,fittings,oilfilters,seals&gaskets,HydraulicOil.Intensifier&accumulators–Applicationareas&Symbols.	**05**
6.	**HydraulicCircuits:** SpeedControllingCircuits-Meterin,Meteroutcircuits&bleedoff Regenerativecircuit,PumpUnloadingCircuit,synchronizationcircuitandSequencingcircuit.	**04**
7.	**Pneumatics:** Compressors–Reciprocating,ScrewandVanetypes,Airreceiver FluidConditioners– Filter,Regulator,Lubricator,FRLunit,Pneumaticsilencer,Airdryer AirControlValves–Checkvalve,shuttlevalve,3-wayand4-waydirectionalcontrolvalve.Flowcontrolvalve,Symbols PneumaticCylinders,PneumaticRotaryActuators.Symbols PneumaticCircuits–Doubleactingcylindercontrol,Controlofanairmotor.	**06**
8.	**ApplicationofFluidPowerinAutomobiles:** 8.1.ApplicationofHydraulics– Powerbrake,powersteering,hydraulicshockabsorber	**04**

	8.2.ApplicationofPneumatics–Airbrake	
	TotalHours	**42**

Weightagedistributioninbothobjective,shortandbroadanswertypequestions:

Group	ModuleNumber	Weightage(%)
A	1,2&3	50
B	4&5	20
C	6,7&8	30

CourseOutcomes:

Attheendofthecourse,thestudentwillbeableto:

CO1	Classifydifferenttypesofpositivedisplacementtypepumpsanddemonstrate theirworkingprinciple.
CO2	Classifydifferenttypes ofcontrol valvesusedinhydraulic&pneumaticcircuits& explaintheirapplication.
CO3	Identifydifferenthydraulicactuatorsandillustratetheirworkingprinciple
CO4	Identifyhydraulic &pneumatic symbols &be able to demonstrate working principleofhydraulic&pneumaticcircuits.
CO5	Identifydifferenthydraulicaccessoriesandexplainapplicationofaccumulator& intensifier.

TextBooks:

1. FluidPowerwithApplications:AnthonyEsposito,Pearson
2. HydraulicandPneumaticControls–K.ShanmugaSundaram,S.Chand
3. Fluid Power: Generation,TransmissionandControl– JagadeeshaT.,ThammaiahGowda,Wiley
4. OilHydraulicSystems:PrinciplesandMaintenance– SRMajumdar,McGrawHillEducation(India)PrivateLimited
5. PneumaticSystems:PrinciplesandMaintenance– SRMajumdar,McGrawHillEducation(India)PrivateLimited
6. HydraulicsandPneumaticsforB.E.PuneUniversity– AvinashG.Patil,VinayakGaikawad,VikashShinde,TechnicalPublications

References

1. FluidPowerEngineering–M.GalalRabie,McGrawHill

Proposed Syllabus of Automotive Electrical & Electronics

CourseCode	AEPC502
CourseTitle	AutomotiveElectrical&Electronics
NumberofCreditsandL-T-P	3[L–3,T–0,P-0]
CourseCategory	PC
Prerequisites	FundamentalknowledgeofElectrical& Electronicstheoriesandapplications.

CourseObjectives:

1. Todiagnoseandrepairthedefectsinthecircuits,toprotectcircuits&understandwork ing ofelectromagneticgaugesaswellaselectricalaccessories.
2. Tounderstandthepurpose,construction,rating,testingofbattery&majorreasonsofbatte ryfailure.
3. Toidentifycomponents,operationandtestingofstartingaswellaschargingsystem.
4. Tounderstandthebasicneed,components,andoperationsofignitionsystemaswellastroub le shootingoftheignitionsystem.
5. Tounderstandlightingsystem&accessories.
6. Totroubleshootvariouscomponentsinelectrical&electronicssystem.

CourseContents:

Module No.	DescriptionofTopic	Contact Hrs.
1.	**Electrical&ElectronicComponents** PurposeandoperationofelectricalcomponentslikeSwitches,relays,sol enoids,buzzers,andresistors. Purposeofcircuitprotectiondeviceslikefuses,maxifuses,circuitbreake rs(Manualandautomaticresettingtypes.)andfusiblelinks Testingofcircuitdefectslikeopencircuit,shorts,shortstogrounds,volta gedrop. WorkingofElectromagneticgaugesliketemperatureGauges,fuelgauge, engineoilpressuregauge,Speedo-metergauge. Featuresofscantester. Workingofelectricalaccessorieslikewindshieldwiper,washerpumps,b lowermotor,electrochromicmirror,powerwindow,powerseat,power doorlock.	**06**
2.	**Battery** Leadacidbattery–construction&operation. ConceptofLowmaintenance,maintenancefree&HybridBattery. Battery–voltage,batteryratingsandbatteryspecifications.	**08**

	Batterytesting– Batteryterminaltest,Leakagetest,SpecificGravityTest,Opencircuitvoltagetest,Highdischargetest&Capacitytest. Batterycharging– Initialchargingprocedure.Slowandfastratechargingandtricklecharging. Batterymaintenanceandsafetyprecautions. ConceptofJumpstarting,it'sprocedure&precaution. Li-ionBatteries-basicconstruction,workingprincipleof LithiumPolymerBattery(LP),**Li-Po** LithiumIronPhosphateBattery(LFP),**LiFePO4** LithiumManganeseOxideBattery(LMO),**LiMn2O4** ComparisonbetweenLead-Acidbattery&Li-ionBattery. Factorsaffectingbatterylife. Batteryfailures– cyclefailure,internalshortcircuit,overcharging,localactionandsulphation.	
3.	**StartingSystem** Purpose,constructionandworkingofstartingsystem. Startingmotortorqueandpowerrequirement. Typesofstartingmotordrive(Bendixandoverrunningclutchtypesonly) - constructionandworking. Testingofstartingsystem– Crankingvoltagetest,Voltagedroptest,Currentdrawtest,startingmotorbenchtest,&no-loadtest. StartingMotortroubleshooting.	**05**
4.	**ChargingSystem** Purposeofchargingsystem. Operationofchargeindicatorlightcircuit. Generalconstruction&operationofautomotivealternator. Initialexcitationandself-excitation. Alternatortesting– Currentoutputtest,Fieldcurrentdrawtest.Regulatoroutputtest CutoutRelay. Regulation-VoltageandcurrentregulationRegulatorforAlternators. AdvantageanddisadvantagesofdynamoandAlternatordrives Chargingsystemtroubleshooting.	**06**
5.	**IgnitionSystem** Purposeofignitionsystem. Classificationofignitionsystems- ContactpointIgnitionSystem&Electronic ignitionsystem. ContactpointIgnitionsystem(BatteryIgnitionandMagnetoignition). Magnetoignitionsystem-constructionandworkingofCDIsystem. Elements and construction of battery oil ignition system: - Ignitioncoiltypes,Distributor,sparkplug,cords,andCondenser,Camangle&Ignitiontiming. Ignition advance &retard timing mechanism-constructionand	**08**

	working. ComparisonbetweenBatteryandMagnetoIgnitionSystem. Electronic(orsolidstate)ignitionsystemwithdistributorcircuitdiagram andworking. ElectronicSparkControl,ElectronicSparkadvance.Comparisonwithconventionalsystem. Distributorless/computer-controlledcoilignitionsystemoperation. SensorsandIgnitionControlModulefortriggeringandtimingofspark. IgnitionSystemTroubleshooting.	
6.	**Lightingsystem&AdvancedElectricalAccessoriesFundamentals** Variouslightingcircuits. VerticalandSlidecontroloflights. Fog light, Slide light, Brake light, indicator lights, andinstrumentlight,reverselight,parkinglight,Trailerlighting,Interiorlighting. ConceptandapplicationofLEDinautomotivelightingsystem. ConstructionandworkingofLEDheadlight,Projectorheadlight(basicconcept) Operationofautomaticheadlightdimming. Operationofautomaticon/offheadlightwithtimedelay. Wiringdiagramofvehicle. Faultsandrectificationofwiringsystem. Operationofkeylessentry. Operationofcommonanti-theftsystem. Purpose&operationofautomaticdoorlocksystem	**05**
7.	**Diagnosisofelectroniccomponents&Systems** Sensor testing: - Oxygen sensor, Engine coolant sensor, Intake airtemp. Sensor, Throttle position sensor, Manifold absolute pressuresensor. FunctionofElectronicControlModule. ElectronicfuelInjectortesting:-onlysoundtest,Ohmmetertest. On-boarddiagnosis(OBD):- Purposeof(on-boarddiagnosticsecondgeneration)OBDII,flashcodesofMalfunctionindicatorlight. OBDIIterminology:-Drivecycle,Trip,Warmupcycle(Definitionsonly). SAEJ2012standardsDiagnosticTroubleCode(DTC):- 5(five)digitsonly. Troublesofelectronicgaugeslike. Gaugereadslowconstantly. Gaugereadshighconstantly. InaccurateGaugereading.	**04**
	TotalHours	**42**

Weight age distribution in both objective, shorthand broad answers type questions:

Group	ModuleNumber	Weightage(%)
A	1&2	30
B	3,4&5	50
C	6,&7	20

CourseOutcomes:

Attheendofthecourse,thestudentwillbeableto:

CO1	Classifydifferenttypesofautomotivebattery&electronicscomponentsandtheir workingprinciple.
CO2	IdentifyStarterMotor&beabletodemonstrateworkingprincipleofstarting system&relatedcircuits.
CO3	IdentifyAlternator&beabletodemonstrateworkingprincipleofchargingsystem &relatedcircuits.
CO4	Classifydifferent components of Ignition system &be able to demonstrate workingprincipleofIgnitionsystem&relatedcircuits.
CO5	IdentifyECM, OBD-II connector &different types of sensors &illustrate their workingprinciple
CO6	IdentifydifferenttypesofDTCfaultcode&troubleshootingmethods

TextBooks:

1. AutomotiveElectricalandElectronics:-AKBABU
2. BasicAutomobileElectricity–C.P.NAKRA
3. AutomotiveElectricalEquipment:-PLKOHLI

e-References

1. https://en.wikipedia.org/wiki/Lithium-ion_battery
2. https://inverted.in/blog/types-of-lithium-ion-batteries-available-in-the-market

ProposedSyllabusofM.V.Act&TransportManagement

CourseCode	AEPC503
	M.V.Act&TransportManagement
NumberofCreditsandL-T-P	3[L–3,T–0,P-0]
CourseCategory	PC
Prerequisites	NIL

CourseObjectives:

1. Toimpartknowledgeofmotorvehicleactinordertoprovidequalityofservice,convenienceof schedulingandeconomics.
2. Toimpartknowledgeofworkingofdifferenttransportorganizations,standardmethodsofre cordkeeping,useofcomputersetc.
3. Toimpartknowledgeonvaluationofvehiclesanddrivingskills.
4. Toimpartknowledgeonvariousautomotiveresearchorganizationsandtheirfunctions.

CourseContents:

Module No.	DescriptionofTopic	Contact Hrs.
01	**MotorVehicleAct** MotorVehicleAct:ShorttitlesusedinMVA,Definitions,TermsregardingVehicl e. LicensingofDriversofMotorVehicle:Necessity,Agelimit,Responsibility of owners, Restriction on holding a driving license,General,PreliminarytestandDrivingtest.Grant,revocationandpowe roflicensingauthority. Conductor's license: Necessity, Eligibility, Documents required andrulesforconductors.Grantandrevocation. Registration of Vehicles: Necessity, Where to be made, How to bemade, Temporary registration, Production of vehicle at the time ofregistration, Certificate of Fitness, Form and manner of display ofRegistration mark, Size of letters and numerals of registration mark,renewal &TransferofOwnershipofMotorVehicle,cancellation/suspensionofOwners hip. Control of Transport: Transport authorities, Difference between STA& RTA, Necessity of Permit, All types of Permits, Transfer of permit,Temporarypermit,Touristpermit,Nationalpermit.Speedlimits.Offe nces,penaltiesandprocedures.Controloftraffic. ConstructionofMotorVehicle:Overalldimensions,Generalprovisionregardi ngconstructionandmaintenanceofmotorvehicle. Road Safety: Road signs, Imposition of Penalties for violation of Actandarticles,Dutiesofdrivers,Dutiesofconductors,Dutiesofhelper. Taxation:Objectives,Basisoftaxationstructurefortwo-wheeler,three-wheeler, goods and passenger vehicles. Methods of levying tax,Taxexemption. Insurance:MotorVehicleInsurance,comprehensive,thirdparty,No-faultliability,Procedureforaccidentclaim.Furnishingofparticulars	10

	ofvehiclesinvolveinaccident.Dutyofdriverincaseofaccidentandinjury. 1.10 Importance of Motor Vehicles (Amendment) Act, 2019:Compensationforroadaccidentvictims,Compulsoryinsurance,Good Samaritans,Recallofvehicles,NationalTransportationPolicy,RoadSafetyBoard,Offencesandpenalties,Taxiaggregators.	
02	**TransportManagement** Comparison of Modes of transport - Road, water, air transport andrailtransport.SelectionofTransportationMode,AdvantagesandlimitationsofRoadTransport. ElementsinTransportManagement: Demand: Law of Demand,Elasticity of Demand, PassengerDemandforTravel. Revenue Function, Bus Fare, Trend in Traffic and RevenueLevel RouteRationalization:NeedforRouteRationalization,ApproachtoRouteRationalization,RationalizingaSingleRoute,AnalyticalSteps. SchedulingofBusService&itsEconomicImplications,Mofussil Service & Scheduling of Services, City Bus Service &Scheduling, Basic Factors in Bus Scheduling, Calculation ofNumberofBuses,CrewScheduling. Traffic Demand, Peak Hour Demand, Limited Stop Services,ReliefServices. Fare,FareforHillyAreas,FaresforDifferentRoutes,Tariffforexpress and Luxury Services, Fares related to Distances, FareforPeakHours Ticketing System, Qualities of a Good Ticketing System, StageandFareStructure. Organization Setup: Government, Semi Government, Private.Setup of STU: Corporate Structure at all levels, Planning ofFleet,MaintenancemanagementofStateTransportUndertaking(STU),Bus RapidTransportsystem(BRTS). Policiesoftransportorganization:Policiestowardspassenger(like Luxury service, concession pass for students and seniorcitizen,Expressserviceonspecialroute,Nightserviceonspecialroute,VehiclesforSpecialprivateTours/Yatras/functions/Educationaltours),Policiestowardsemployees(Bonus, Overtime Allowances, Health Insurance, free TravelPackagesEmployeeTrainings). Scheduling:Basicfactorsinbus,crew(staff)andmaintenancescheduling,calculationofnumberofbuses. Freight Calculation: Time base, Distance base, Contract, perpassenger,cubicfeet tonemethod.Structure of fare,fixedcost, Maintenance cost, depreciation cost, insurance, intereston capital, variable cost, Hiring of trucks, Toll, staff wages,Miscellaneouscost. Recordkeeping:Logbook,Tripoperationalsheet,Vehicleledger, Truck history card, Monthly operational sheet, Goodsconsignment note, daily fuel consumption, various types ofbookings,UseofComputer. GoodsTransportOperation:Schedulingofgoodstransport, ManagementInformationSystem(MIS)inpassenger/goods	20

	transportoperation,Storage&transportationofpetroleumproducts, Intelligent Transport System (ITS), Traffic navigation,Globalpositioningsystem.	
03	**EstimationandValuationofVehicle** Roleofsurveyor,Procedureofsurveyandvaluationofvehicle,Accidentsurvey report. Importance of warranty system and protection of law: How to dealwithdefects,benefitsofwarrantysystem.Protectionoflaw. Buying a new vehicle: Factors to be considered - Ex-showroom priceandonroadprice,useofvehicle,whenandwheretobuy,comparisonofdifferentspecifications. Buying a used vehicle: When & where to buy: Dealers, used car firms,Privatesellers,Garages,Auctions.Factorstobeconsideredwhilebuying ausedvehicleascustomerpointofview- Depreciation,Modelandyear,Checkingofmechanicalandelectricalaccessories,Testdrive. Preparations for selling a vehicle from seller point of view: When tosell, How to sell, Auctions, Garages, Private sale, Preparing the car,Documentation,Sellingprice,Safeguards.	08
04	**AutomobileIndustry,it'sFunctions&Role** DevelopmentofmotorindustryinIndia,Structureofautomobileindustry,ImportanceofAutomobileEngineer. Workingofvarioustypesoftransportorganizations:CRRI,PCRA,CIRT,ARAI,VRDE Workingof Various State Transport Organizations: CSTC, SBSTC, WBSTC,NBSTC,CTC,WBTC	04
TotalHours		**42Hrs**

Weightagedistributioninbothobjective,shortandbroadanswertypequestions:

Group	ModuleNumber	Weightage(%)
A	1	30
B	2	40
C	3&4	30

CourseOutcomes:

Attheendofthecourse,thestudentwillbeableto:

CO1	InterprettheMotorVehicleActandTrafficRules.
CO2	ApplytheconceptofTaxationandInsuranceinvehicleregistration.
CO3	Illustratethedifferentaspectsrelatedtotransportationsystem.
CO4	Identifytherolesofassessorandsurveyorinautomobileindustry.
CO5	Understandtheroleofvariousresearchorganizationinmotorindustry.

TextBooks:

1. VehicleTransportManagement,S.L.Bhandarkar,DhanpatRai&Co.
2. TransportManagement,R.B.Gupta,SatyaPrakashan.

3. MotorVehicleActandTransportmanagement,V.S.Khilery,SatpalSharma,ShamanGupta,IshanPublications.

ReferenceBooks:

1. MotorVehicleAct,1988-GovernmentofIndiaPublications
2. CentralMotorVehiclesRules,1989-GovernmentofIndiaPublications
3. MotorVehicles(Amendment)Act,2019-GovernmentofIndiaPublications
4. ManagementofStateRoadTransportsinIndia,D.K.Kulshrestha,MittalPublications
5. RuralTransportinIndia,K.N.Ramanujam,MittalPublications
6. IndianTransportSystem,P.JegadishGandhiandG.JohnGunaseelan,MittalPublications
7. TransportManagementinIndia,D.K.Kulshrestha,MittalPublications

ModelNumerical(Suggestiveonlyfor2.2.10)

1. CalculatenumberofbusesrequiredfortransportoperationbetweenterminalsKALNAtoASANSOLandprepareabusscheduleforthedatagivenbelow-

(i) DistancebetweenKALNAtoASANSOLis169km

(ii) Busesareoperatedthroughthedaybetween4a.m.to11p.m.withfrequencyof2Hr.Runningtimeis5Hrs. andstandingtimeis3Hrs.

Soln.:

CalculationofNumberofBusesbetweenKALNAtoASANSOLBusschedulingdataaregivenbelow:
DistancebetweenKalnatoAsansolis169km.
Busesareoperatedthroughthedaybetween4a.m.to11p.m.withfrequencyof2Hr.Runningtimeis5Hrs. andstandingtimeis3Hrs.

Byusingbelowgivenformulawehavetocalculatethenumberofbusesrequiredfortheschedule.

$$N=\frac{(ST+RT)}{2}F$$

where,N=Numberofbuses
ST = Stand/ Terminus
timeRT=Running/Journeytime
F=Frequency

Numberofbusesrequired=8(followingtheequation)

1STBUS			2NDBUS		
	STATION	**TIME**		**STATION**	**TIME**
DEPARTURE	KALNA	4am	DEPARTURE	ASANSOL	4am
ARRIVAL	ASANSOL	9am	ARRIVAL	KALNA	9am
DEPARTURE	ASANSOL	12 pm(noon)	DEPARTURE	KALNA	12pm(noon)

ARRIVAL	KALNA	5pm	ARRIVAL	ASANSOL	5pm
DEPARTURE	KALNA	8pm	DEPARTURE	ASANSOL	8pm
ARRIVAL	ASANSOL	1am	ARRIVAL	KALNA	1am
3RDBUS			**4THBUS**		
	STATION	**TIME**		**STATION**	**TIME**
DEPARTURE	KALNA	6am	DEPARTURE	ASANSOL	6am
ARRIVAL	ASANSOL	11am	ARRIVAL	KALNA	11am
DEPARTURE	ASANSOL	2pm	DEPARTURE	KALNA	2pm
ARRIVAL	KALNA	7pm	ARRIVAL	ASANSOL	7pm
DEPARTURE	KALNA	10pm	DEPARTURE	ASANSOL	10pm
ARRIVAL	ASANSOL	3am	ARRIVAL	KALNA	3am
5THBUS			**6THBUS**		
	STATION	**TIME**		**STATION**	**TIME**
DEPARTURE	KALNA	8am	DEPARTURE	ASANSOL	8am
ARRIVAL	ASANSOL	1Pm	ARRIVAL	KALNA	1Pm
DEPARTURE	ASANSOL	4pm	DEPARTURE	KALNA	4pm
ARRIVAL	KALNA	9pm	ARRIVAL	ASANSOL	9pm
DEPARTURE	KALNA	12am	DEPARTURE	ASANSOL	12am
ARRIVAL	ASANSOL	5am	ARRIVAL	KALNA	5am
7THBUS			**8THBUS**		
	STATION	**TIME**		**STATION**	**TIME**
DEPARTURE	KALNA	10am	DEPARTURE	ASANSOL	10am
ARRIVAL	ASANSOL	3pm	ARRIVAL	KALNA	3pm
DEPARTURE	ASANSOL	6pm	DEPARTURE	KALNA	6pm
ARRIVAL	KALNA	11pm	ARRIVAL	ASANSOL	11pm
DEPARTURE	KALNA	2am	DEPARTURE	ASANSOL	2am
ARRIVAL	ASANSOL	7am	ARRIVAL	KALNA	7am

Proposed Syllabus of Automotive Electrical & Electronics Lab

Course Code	AEPC504
Course Title	Automotive Electrical & Electronics Lab
Number of Credits and L-T-P	1[L–0,T–0,P-2]
Course Category	PC
Prerequisites	Nil

Course Objectives:

1. Tounderstandvarioustestproceduresforbatteryasspecifiedbymanufacturerandtheprecautionswhilehandlingabattery.
2. Toidentifythealternatorcomponents,startermotorcomponentsandunderstandtestprocedureofsame.
3. Toknowthetestandserviceprocedureforsparkplug,distributorandsparkplugcords.
4. ToIdentifyandlocatesensorsandtounderstanddiagnosticprocedures(on-boardandstandalonediagnosis).
5. To identify the vehicle wiring system.

Course Contents:

Exp. No.	Description of Experiments	Contact hours
1.	1.1 Study, testing (Specific gravity of electrolyte, High rate discharge test of battery & Load test of battery), sketching of constructional details, and working principle of battery. OR	4
	1.2 Study, testing (Nominal Cell Voltage, Discharge end voltage or cut offvoltage,RatedCapacity,Weightofthecell,ChargingVoltage,InternalResistance,EnergyDensity(wh/kg))andsketchingofconstructional Details and working principle of battery.	4
2.	2.1 Study, testing and sketching of various components and the functions of coil ignition systems and Inspection of spark plug cords, Servicing of spark plugs and distributor. OR	4
	2.2Study,testing&sketchingof various components and their function of Electronic Ignition system.	4
3.	3.1Study,testing and sketching of starting system and the constructional details of self-starter. OR	4
	3.2StarterMotor–componentidentification, starter current draw test and voltage drop test.	4
4.	4.1Study, testing and sketching of charging system and the constructional details of Automotive Alternator. OR	4
	4.2 Alternator-component identification and output test, Regulated	4

	VoltageOutputTest,chargingcircuitresistancetest.Electricaltestingof Rotor and stator of alternator.	
5.	5.1Study, testing and sketching of Headlight Circuit with upper and dipper function. OR	4
	5.2Study, sketching and working principle of Automatic Headlight Dimming Circuit	4
6.	6.1Study, testing and sketching the wiring diagram of Electronic Control Module (ECM or ECU) with different sensors and actuators. OR	4
	6.2Study,testingandfaultcodeterminologyandrectificationprocessofOBD-II	4
7.	7.1Study,testingandsketchingofVehicleWiringSystem OR	4
	7.2 Study, location and identification of different sensors.	4
	Total Hours	**28**

Course Out comes:

Attend of the course ,students will be able to:

CO1	IdentifybasictoolsofAutomotiveElectricalandElectronicsLaboratory
CO2	Identifyanddiagnosedifferenttypesofcomponentsandfaultsinautomotive ignitionsystem
CO3	Identifyanddiagnosedifferenttypesofcomponentsandfaultsinautomotive Startingsystem
CO4	Identifyanddiagnosedifferenttypesofcomponentsandfaultsinautomotive Chargingsystem
CO5	IdentifyECU/ECMandOBD-IIalsotheirwiringandworkingprocedure

ReferenceBooks:

1. AK.BABU.AutomotiveElectricalandElectronics,KhannaPublishing
2. Dr.KirpalSingh,AutomobileEngineering(Vol.–1&2),StandardPublishersDistributors,NewDelhi.
3. P.L.Kohli,AutomotiveElectrical,TataMcGrawHillEducation,NewDelhi.
4. C.P.NAKRA,BasicAutomobileElectricity,DhanpatRaiPublishingCompany,

ProposedSyllabusof Modern Manufacturing Methods

CourseCode	AEPE511
CourseTitle	ModernManufacturingMethods
NumberofCreditsandL-T-P	3(L-3,T-0,P-0)
CourseCategory	PE
Prerequisites	Nil

CourseObjectives:

1. ToacquirebasicknowledgeaboutModernManufacturingMethods.
2. To gain knowledge about different types of nontraditional machining processesalongwiththeircharacteristicsandproperuseinautomotiveindustries
3. TogainknowledgeaboutconceptsofconstructionalfeaturesofCNCmachinetools along with programming method, advantages, limitations and applicationsrequiredforautomotiveindustries
4. Toacquirebasicconceptofadditivemanufacturingusedinautomobilesector.
5. Togatherbasicknowledgeaboutsomemanufacturingprocessesverymuchneedfulinau to motive sectors liked ie casting, spotwelding, seamwelding

CourseContents:

Module No.	Description of Topic	Contact Hrs.
1	1. Introduction to Modern Manufacturing Process Concept of jobproduction,batchproduction,massproduction&continuousproducti onandtheirapplicationinautomobileindustries Development of machinetools(onlyelementaryidea)– generalpurposemachinetools,semiautomaticmachinetools,automatic machinetools;applicationsinautomobileindustries Needs of Modern Manufacturing Processes Pneumatic & hydraulic clamping devices used in modern shop Concept of jig & fixture; differences between them, use of Fixture indifferent types of manufacturing processes	**(6)**
3	2. Joining Processes Definition, purpose and types of joining processes Welding: Concept, types with examples Arc welding: concept, types with application Resistance welding: working principle, types with applications Brazing & soldering: Concept & applications	**(10)**
4	3. Non Traditional Machining Definition and purpose of non-traditional processes Classification of non-traditional processes with suitable	**(10)**

	Example and application Working principle, advantages, limitations and application of: i) Ultrasonic machining ii) LASER beam machining iii) EDM & EDM wire cut iv) Electro Chemical Machining	
4	4. Additive Manufacturing: Concept, Advantages ,Disadvantages, Applications	**(2)**
5	5. Computer Numerical Control Machining Difference between NC&CNC with respect to advantages, limitations and application Different components of CNC machine tools (only function and application):Stepper motor,Servomotor,Encoders(rotary&linearencoder),Recirculatingballscrew,Automatictoolchanger,Toolmagazine. Workholdingdeviceforturningandmachiningcentres(onlynameandrelativeadvantages,limitations) Manualpartprogramming:referencepoint(Machine Zero, Program Zero, Part Origin), Axis identification of Turning Centre & Machining Centre, CNC Codes form annual part programming G–codes, M- Codes, Spindle speed control, feed rate control, Tool selection)part programming for turning centre using different codes & fixed cycles for obtaining step, taper, plain & circular turning, facing, external threading & parting off operation.	**(14**
	TotalNumberofContactHours	**42**

Weightagedistributioninbothobjective,shortandlonganswertypequestions:

GroupName	UnitNumber	Weightage(%)
A	1,2,3	60
B	4,5,	40

CourseOutcomes:

Attheendofthecourse,thestudentwillbeableto:

CO1	Illustratefundamentalconceptsonmodernmanufacturingmethodsalongwithconceptsonjig&fixtureandtheirapplicationinautomotive industries
CO2	Statefundamental concept on different joining process, their types, applicationsalongwithbriefconceptofArcandResistancewelding.
CO3	Explainworkingprinciples,advantages,limitationsandapplicationsofsomenon-traditionalmanufacturingprocessessuchasUSM,LBM,EDM& ECMalongwithbroadclassificationofnon-traditionalprocesses.

CO4	Statefundamental concept on Additive Manufacturing Process, their advantages,disadvantagesandapplications
CO5	ExplainfunctionofdifferentpartsofaCNCmachinetoolsalongwith demonstrationofmanualpartprogrammingofCNCtoobtainbasicmachiningoperationsrequiredinautomotiveindustries

ReferenceBooks:

1. ManufacrutingTechnologyMetalCutting&Machnetools-P.N.Rao,TataMcGraw-Hill
2. CNCmachines-PablaB.S.M.Adithan,Newageinternationallimited
3. NonconventionalMachining-P.K.Mistra,NarvasaPublishiningHouse
4. ElementsofworkshopTechnology–VolumeI&II-S.K.HajraChaudary,Bose,Roy,MediaPromotersandPublisherslimited
5. WorkshopTechnologyVolume-I&II-H.S.Bawa,TataMcGraw-Hill
6. ManufacturingEngineeringandTechnology(SIEdition)-SeropeKalpakjianandStevenR.Schmid, Pearson Publication
7. ToolEngineering-BrentDonaldson,**Tata**McGraw-Hill

ProposedSyllabusofAutomobileAirConditioning

CourseCode	AEPE512
CourseTitle	AutomobileAirConditioning
NumberofCreditsandL-T-P	3[L–3,T–0,P-0]
CourseCategory	PE
Prerequisites	HeatPowerEngineering,

CourseObjectives:

1. TounderstandthenecessityofAirConditioningincar.
2. TounderstandworkingprocessofcarAirConditioningsystem.
3. TounderstandcomponentofcarAirConditioningsystemandthereuses.
4. TorecognisetheprocessofAirConditioningunitrepairandmaintenance.
5. TorecogniseenvironmentalaspectsrelatedtoAirConditioningSystems.

CourseContents:

Module No.	DescriptionofTopic	Contact Hrs.
1.	**IntroductionofAirConditioningsystem** WhatisAirConditioningsystem–UsagesofACsystem-TypeofAirConditioningsystem(onlyWorkingandConstruction) Refrigerant-PropertyofRefrigerant–Eco-friendlyRefrigerant CarRefrigerant–R12–R134a–R1234yf-Property-Advantages–Disadvantages HumanComfortControl–ComfortZone–airmovement–windchillfactor–odourproblem –humidity–safety. RequirementsofAirconditionsystem–Heat–Ventilation–humidity	**05**
2	**CarAirConditioningsystemanditsComponents** Layout of Car Air Conditioning system – Different Category of Car airconditioningsystem. Construction&workingoffollowingrefrigerationsubsystems–thermostatic expansion valve, fixed orifice tube & rotary vane aircyclesystem. Construction & working of evaporator - condenser - accumulator - aspirator Receiverdriers&accumulator-Types,construction&Working ReciprocatingCompressor,scroll&rotaryvanecompressors(Construction& working)-Drivesystemsforcompressors-CompressorClutches-CompressorClutchelectricalcircuit-Compressorlubrication. Construction& working ofvacuum operateddevices i.e., vacuum reservetank,vacuumrestrictor,vacuummotor,checkvalve&check	**12**

	relays. Construction&workingofelectromagneticclutch. Meteringdevices- comparisonofthermostaticexpansionvalve&fixedorificetube. Types working & comparison of thermostatic expansion Valves i.e., Hvalve,blocktype,internallyequalized&externallyequalized. Functionsofthermostaticexpansionvalve- Throttlingaction,modulatingaction&controllingaction. DifferencebetweenconventionalAirConditioningsystemandCar AirConditioningsystem	
3.	**Air-conditioningunitinletanddeliverysystem** Construction&workingofAirintakesection,coresection&distributionsection Construction&workingofDownstream–upstream-split-hybridCaseandductsystem. Construction&workingofrearheating&coolingsystem Hoses&connectors– constructionofsystemhoses,charginghosewithshutoffvalve& connectors.	**04**
4	**AirConditioningsystemsensorsandcontrol** System controls – Construction & working of typical vacuum system&electronictemperaturecontrolsystem. Switches – Construction & working of high side temperature switch,low-side temperature switch, high pressure switch, low-pressureswitch,pressureregulator,ambientswitch&superheatswitch. Sensors- Construction&workingofsunloadsensor,outsidetemperaturesensor&incar temperaturesensors. Differenttypesofsensorsandactuatorsusedinautomatictemperature control - Fixed and variable displacement temperaturecontrol- Semi Automatic - Controller design for Fixed and variabledisplacementtypeairconditioningsystem Construction & working of blower clutch control, heater control, andtimedelayrelayforheatercontrol. Modedoorsandtemperaturedoors–definition–application–working–difference. Electricalcircuits-Typicalclimatecontrolsystem&Electronicclimate controlsystem,theirelectricalcircuits&working	**09**
5	**MaintenanceofCarAirConditioningSystem** Visual & acoustic check - side glass - leak test - temperature test - Procedureofcharging&discharging-Moistureremovalprocedure. Service equipment & tools – Vacuum pump - Manifold & gauge i.e.,Low side & high side - gauge calibration - recovery unit - recyclingunit-Halide(2reon)-Fluorescentleakdetector-nitrogenleaktest. Compressorservice–Symptoms,faults,cause&remedy.	**12**

	Electromagneticclutchservice–Symptoms,faults,cause&remedy. Performance testing procedure of thermostatic expansion valve &fixedorificetube. Refrigerant-Packaging, storage,restrictions,colour code& puritytest. Common troubleshoot - No Cold Air from A/C - No Air Coming fromtheVents-AirisCoolbutDoesNotGetCold-ACSmellsLikeMildew-CarACMakesNoiseWhenTurnedOn-WaterontheFloorboards-AirConditioningGoesfromColdtoHot	
	TotalHours	**42**

Weightagedistributioninbothobjective,shortandbroadanswertypequestions:

Group	ModuleNumber	Weightage(%)
A	1&2	45
B	3&4	25
C	5	30

CourseOutcomes:

Attheendofthecourse,thestudentwillbeableto:

CO1	IdentifythedifferentcomponentofcarAirConditioningsystem
CO2	IdentifydifferentSensorsystemandtheirusesinCarAirConditioningsystem
CO3	DifferentiatetheconventionalAirConditioningsystemwithcarAirConditioning system.
CO4	RepairerandMaintenancecarAirConditioningunit.
CO5	IdentifycommonproblemofCarAirConditioningsystemandtheirsolution.

TextBook:

1. AutomotiveAirconditioning:WilliamHCrouseandDonaldLAnglin,McGrawHillInc., 1990.
2. AutomotiveHeating,Ventilation,andAirConditioningsystems:WarrenFarnellandJamesD.Halderman,ShopManual,PearsonPrenticeHall,2004
3. AutomobileEngineering(Volume–VI):AnilChhikara,SatyaPrakashan.
4. AutomotiveAir-conditioning:CliffordL.Samuels,PrenticeHallInt.
5. AutomotiveAirconditioning&climatecontrolsystem:StevenDaly,CengagePublication

ReferenceBooks:

1. AutomotiveAirConditioning:Goings,L.F.,AmericanTechnicalservices,1974.
2. MitchellAutomaticHeatingandAirConditioningSystemsMitchellInformationServices,Inc.,PrenticeHallInc.,1989.
3. PaulWeisler,"AutomotiveAirConditioing",RestonPublishingCo.Inc.,1990.
4. U-tubevideorelatedtoA/Cmaintenance.

ProposedSyllabusofEarthMovingEquipments&FarmMachinery

CourseCode	AEPE513
	EarthMovingEquipments&FarmMachinery
NumberofCreditsandL-T-P	3[L–3,T–0,P-0]
CourseCategory	PC
Prerequisites	EngineeringMechanics,AutomotiveChassis, AutomotiveEngine,AutomotivePowertrain,Fluid Power

CourseObjectives:

1. Toimpartknowledgeontheimportanceofearthmovingmachines&agriculturalmachinesin India.
2. Toidentifyvarioussystems&subsystemsofearthmovingmachines&agriculturalmachines
3. Toexplainworking&constructionofvarioussystems&subsystemsinearthmovingmachine s&agriculturalmachines
4. Tocarryoutpreventivemaintenanceofearthmovingmachines&agriculturalmachines.

CourseContents:

Module No.	Description of Topic	Contact Hrs.
01	**Earth Moving Equipments–Introduction** Types ofearthmovingequipmentsandoutlinedtheiruses. GeneralspecificationsofatypicalEarthMovingMachine. ComparisonbetweenGeneralAutomobileandEarthMovingMachineonf ollowingparameters: a) TravelingSpeed b) Workingconditions c) Poweroutput&powervariations d) EngineandPowertrain(basicconcept) e) PneumaticControls f) Torque&torquevariations. g) Steering h) Suspension i) Fuel&fuelconsumption j) Powertakeoffs k) Clutch l) Brakes m) Drivinglicense n) RTOregistration	**10**
02	**Tractors:** Comparisonoftractorwithanautomobile. GeneralLayoutofatractor. ComparisonbetweentyredandCrawlerTractor. Powertrain&transmissionlayoutofatractor.	**08**

	Tractorpowertakeoffitsworking&construction. Counterweight&itsimportance. Factorsofselectionoftractors&uses. Troubleshooting,care&maintenance. Powertiller-Comparisonwithtractors.	
03	**Differentotherearthmovingequipments&farmmachineries***[functions, working principle, detail constructional features (with powerflowlayout&specialfeatures),usesandtroubleshooting]***:** a) Dozer b) DumpTrucks c) Loader d) Roller e) Shovel f) Cranes g) ForkLift h) Scrapers i) Rippers j) Excavator k) Tanker/Trailer/Carrier	**20**
04	**PlantProtectionEquipments&Harvestingequipments:** SprayerandDuster(types)– Construction,Function,workingprinciple&applications. Mowers,Reapers, Binders, Forge Harvester, Vegetable and Food Harvester–Construction,Function,workingprinciple&applications.	**04**
TotalHours		**42Hrs**

Weightagedistributioninbothobjective,shortandbroadanswertypequestions:

Group	ModuleNumber	Weightage(%)
A	1	30
B	2	40
C	3&4	30

CourseOutcomes:

Attheendofthecourse,thestudentwillbeableto:

CO1	Understandthetypes,special features,workingprinciple,applicationofvarious EarthMovingEquipments
CO2	Explainthe purpose, constructional features and operation of agriculture tractors
CO3	Identifyvarioussystems&subsystemsofearthmovingmachines&agricultural machines
CO4	Carryout preventive maintenance of earth moving machines &agricultural machines

TextBooks:

1. LatestDevelopmentofHeavyEarthMovingMachinery,AmitoshDe,GalgotiaPublicationsPvtLtd,NewDelhi
2. HeavyEarthMovingMachinery,AmitoshDe,LovelyPrakashan,Dhanbad
3. FarmMachinesandEquipment,C.P.Nakra,DhanpatRaiPublishingCompany(P)Ltd.

ReferenceBooks:

1. AgriculturalTractorandMachinery,D.N.Sharma,S.MukeshandVivekBalyan,JainBrothers, NewDelhi
2. FarmTractor:MaintenanceandRepair,S.C.Jain,StandardPublishes-Distributors
3. FarmMachinery-AnApproach,S.C.Jain,StandardPublishes-Distributors
4. FarmPower&Machinery,ErSanjayKumar,KalyaniPublications

ProposedSyllabusofTwo&ThreeWheelers

CourseCode	AEPE514
	Two&ThreeWheelers
NumberofCreditsandL-T-P	3[L–3,T–0,P-0]
CourseCategory	PE
Prerequisites	AutomotiveEnginesandAutomotiveChassis

CourseObjectives:

1. Toimpartknowledgeofdifferenttypesoftwoandthreewheelers.
2. Toimpartadequateknowledgeonfameandbodyoftwoandthreewheelers.
3. Toimpartknowledgeonpowerunit,varioussubsystemsoftwoandthreewheelers.
4. Toimpartknowledgeofpropermaintenanceoftwoandthreewheelers.

CourseContents:

Module No.	DescriptionofTopic	Contact Hrs.
01	**Introduction** Classification of Two Wheelers: Types- Mopeds,Scooters,Motorcycles. BasicSystems-Frame, WheelsandBrakes,Suspensions,Engine,DriveTrain,FuelSystem,Ignition,ElectricalSystem(briefideasonly) ClassificationofThreeWheelers:Withrespecttouse,make,fuelused,wheelconfigurationandsteeringconfiguration. Typesof Loading Auto Rickshaws (layout) - Passenger, Delivery, Pickup,Tippervehicle.	**04**
02	**FrameandChassis** LayoutofMoped,Scooter,Motorcycle. LoadontheFrame,DesignConsiderations,ComponentsofFrame,MountingProvisionsonFrame TypesofFrames: TubularFrame-Backbone,DoubleBeam,SingleCradle,DoubleCradle,Trellis,Step-ThroughFrame. EngineBasedFrame Twin-SparFrame MonocoqueFrame VibrationDampers-HandlebarMountings,Handlebar,SteeringDampers FrameMaterials BodyWorks ErgonomicConsiderations Sidecars	**06**

03	**PowerUnitForTwo&ThreeWheelers** EnginesforTwoWheelers ArrangementsofCylinders-singlecylinder,Inline(2,3,4cylinder),VTwin,VFour,HorizontallyOpposedTwin. Valve Operating Mechanism - Pushrod Overhead Valve, SingleOverheadCamshaft,DoubleOverheadCamshaft. CrankingSystem:BasicCrankingMechanism-Rollertyperatchet, Lock pawl type ratchet, Regular ratchet wheel. PushStarting, Moped Cranking Mechanism - Indirect, Direct. KickStartMechanism,AutoStartMechanism. Exhaust System of Two Wheelers, Exhaust Pipe and Header -Two into two, Two into one, Three into one, Four into four,Fourintotwo,Fourintoone.Muffler EnginesforThreeWheelers FourStrokeCNG,FourStrokeDieselEngine	**12**
04	**Sub-SystemsforTwo&ThreeWheelers** TransmissionSystemofTwo&ThreeWheeler LayoutofTransmissionSysteminTwoWheelers,PrimaryReduction - Belt Drive, Chain Drive, Gear Drive. Comparison ofDrives. GearShiftingMechanismofTwo&ThreeWheeler-HandOperated,FootOperatedShiftingMechanism. FinalDrive-BeltDrive,ChainDrive,ShaftDrive. DriveTrainLayoutofAutoRickshaw. SteeringSystemofTwo&ThreeWheeler SteeringGeometryandEffects,SteeringColumnConstruction HandleBar-TypesandConstruction,ApeHanger,Beach,Drag,Clip-onsHandlebar. HandlebarControls- RightHandlebar,LeftHandlebar.Controlsonfoot. SuspensionSystemofTwo&ThreeWheeler SuspensionRequirements-Kinematic,Dynamic. DesignConsiderations SpringerFork,GirderFork,Trailing&LeadingLinktypeSuspension, Single&DoubleLinkTypeFrontSuspension,Hardtail&SwingArmTypeRearSuspension,UpsideDownFork. RearSuspensionSystemofPassenger&LoadingAutoRickshaw. BrakingSystemofTwo&ThreeWheeler HandOperatedand FootOperatedMechanicalBrake,HandOperated and Foot Operated Hydraulic Brake, Antilock BrakingSystem-ComponentsandOperation. BrakingSystemofAutoRickshaw.	**16**

05	**ServiceandMaintenance** Servicingandmaintenanceofcarburetor,sparkplugcleaning,ignition timing adjustment, gear box, steering head, shaft and chaindrive,suspension,brake servicingandadjustment. Periodicmaintenanceschedules. Fact and Features of Two wheelers: Power to Weight Ratio, BasicCalculation,DifferentStandsforTwowheelers,Differentsizeoftyresofatwowheeler.	**04**
TotalHours		**42Hrs**

Weightagedistributioninbothobjective,shortandbroadanswertypequestions:

Group	ModuleNumber	Weightage(%)
A	1&2	25
B	3	35
C	4&5	40

CourseOutcomes:

Attheendofthecourse,thestudentwillbeableto:

CO1	Identifydifferenttypesoftwowheelersandthreewheelers.
CO2	Explaintheworking,componentsandaccessoriesoftwoandthreewheeler powerunits.
CO3	Describethetransmissionandcontrolsystemoftwoandthreewheelers.
CO4	Classifybrakesandsuspensionsystemoftwoandthreewheelers.
CO5	Explaintheprocedureforservicingandmaintenanceofdifferentcomponents usedintwoandthreewheelers.

TextBooks:

1. Twoand ThreeWheelerTechnology, Dhruv U. Panchal, JayeshP.Maniar and JigarA.Doshi,PHILearning.
2. TwoWheelersandThreeWheelers,K.K.Ramalingam,ScitechPublications(India)Pvt.Ltd.

ReferenceBooks:

1. S.Srinivasan,AutomotiveMechanics,TataMcGrawHillEducation,NewDelhi.
2. S.K.Gupta,AutomobileEngineering,S.Chand,NewDelhi.
3. R.B.Gupta,AutomobileEngineering,SatyaPrakashan,NewDelhi.

ProposedSyllabusofInternship–II

CourseCode	SI501
CourseTitle	Internship-II
NumberofCreditsandL-T-P	1[L–0,T–0,P-0]
CourseCategory	Internship
Prerequisites	Upto4thsemesterallsubjectknowledge

CourseObjectives:

1. ExplorecareeralternativespriortoDiploma.
2. Integratetheoryandpractice.
3. Assessinterestsandabilitiesintheirfieldofstudy.
4. Learntoappreciateworkanditsfunctionintheeconomy.
5. Developworkhabitsandattitudesnecessaryforjobsuccess.
6. Developcommunication,interpersonalandothercriticalskillsinthejob interviewprocess.
7. Buildarecordofworkexperience.

After 4thSemester, for Internship-II, students are to be involved to undergo internship withindustry/ NGO's/ Government Organizations/ Centre of Excellence/ Micro/ Small/ MediumEnterprisestomakethemselvesreadyfortheindustry.InthiscasealsoTrainingandPlacement Cell and concerned teachers have to take initiative for coordination. Online internship optionsmaybeexplored.

For more guidance regarding internship, refer AICTE Internship Policy and AICTE InternshipPortal(www.internship.aicte-india.org).

After completion of Internship, the student should prepare a comprehensive report to indicatewhat he/ she has observed and learnt in training period. The student may contact IndustrialSupervisor / Faculty Mentor / TPO for assigning topics and problems and should prepare thefinal report on the assigned topics. The training report should be signed by the IndustrialSupervisor/InternshipFaculty Mentor,TPOandHOD.

TheInternshipReportwillbeevaluatedonthebasisoffollowingcriteria(asapplicable)

60Marks

Sl.No.	CriteriaforevaluationofInternshipReport
1	Originality
2	Adequacyandpurposefulwriteup
3	WritingSkill(Organizations,format,drawing,sketches,style,language)
4	Attendancerecord,dailydiary,qualityofInternshipReport.

Seminarsmustbearrangedforthestudentbasedonhis/hertrainingreport,beforeanInternalCommittee constituted by the concerned department of the Institute. The evaluation will bebasedonthefollowingcriteria:

40Marks

Sl.No.	CriteriaforevaluationofInternshipReport
1	Qualityofcontentpresented
2	ProperPlanningforpresentation
3	Effectivenessofpresentation
4	Depthofknowledgeandskills
5	Vivavoce

CourseOutcome:

Attheendofthecourse,thestudentwillbeableto:

CO1	Learntheapplicationofknowledgeinrealworldproblems
CO2	Getexposuretoteam-workandleadershipquality.
CO3	Dealwithindustry-professionalsandethicalissuesintheworkenvironment

ProposedSyllabusofMajorProject

CourseCode	PR502
CourseTitle	MajorProject-I
NumberofCreditsandL-T-P	1[L–0,T–0,P-2]
CourseCategory	MajorProject
Prerequisites	AllsubjectRelatedtoAutomobileEngineering

CourseObjectives:

1. Tounderstandthesolutionprocessofreal-lifeproblem
2. Toachievethepotentialityofdoingteamwork
3. Tounderstandthe gapbetweenacademicknowledge andactualreal-lifeproblem-solvingknowledge.
4. Topreparetheprojectrepotinaprofessionalmanner.

ProjectGroup:

1. Formationofprojectgroup:Maximum8to12studentsperbatch.
2. Eachprojectgroupshouldselectworkbyconsultingtheguide.

CourseContent:

Itisthe1st partofthemajorproject.Duringthismajorproject-I,partialwork *(Reviewofrelatedliterature,MarketSurveyetc.*)ofthefinalprojecthavetobe completed.

Theprojecttopichasbeenselectedonanadvancedtopicof*AutomobileEngineering*.

Note:
Faculty Members and students can select any project topic through mutual discussion for each student groups but related to automobile engineering.

Aftercompletionofthemajorproject-I,eachstudentsgroupshouldprepareonecomprehensivereporttoindicatewhatareobserved,learnt duringtheprojectwork,estimation(timeandcost),work plan, layout of the project during this work. The student may contact guide for assigningtopic and problem and should prepare the progress report on the assigned topics. The projectreportshouldbesignedbytheguideandHOD.

Theformatoftheprogressreportwillbeasfollowing

Sl.No.	Particulates
1	Titlepage
2	Certificatefromguide
3	Abstract
4	Objective
5	Literaturereview/backgroundsurvey/history

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)

Syllabus
of

Diploma in Automobile Engineering [AE]

Part-III (6th Semester)

2023

Semester-wise Detailed Curriculum

Semester VI (Third year)

Branch/Course: Automobile Engineering

Sl. No.	Category	Code	Course Title	Hours per week			Total contact hours/ week	Credits
				Lecture	Tutorial	Practical		
1	Program Core Course	AEPC 601	Garage Practice	3	0	0	3	3
2	Humanities and Social Science Course	HS302	Entrepreneurship and Start-ups	2	1	0	3	3
3	Open Elective	AEOE 621	Engineering Economics and Project Management	3	0	0	3	3
4	Open Elective	AEOE 622/ 623/ 624/ 625	Any one Open Elective	3	0	0	3	3
5	Program Elective Course	AEPE 611/ 612	Any one Programme Elective	3	0	0	3	3
6	Program Core Course	AEPC 602	Garage Practice Lab	0	0	2	2	1
7	Program Core Course	AEPC 603	Driving Practice Lab	0	0	2	2	1
8	Major Project	PR602		0	0	6	6	3
9	Seminar	SE602		0	0	4	2	2
Total				**1**4	**1**	**1**4	**2**9	**22**

List of Programme Electives for Sixth Semester

1. AEPE 611: Modern Vehicle Technology
2. AEPE 612: Vehicle Aerodynamics & Design
3. AEPE 613: Vehicle Body Engineering

List of Open Electives for Sixth Semester

1. AEOE 621: Engineering Economics and Project Management
2. AEOE 622: Electric Vehicle Technology
3. AEOE 623: Industrial Management
4. AEOE 624: Industrial Safety
5. AEOE 625: Disaster Management

Proposed Syllabus of Garage Practice

Course Code	AEPC601
	Garage Practice
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	Automotive Engines and Automotive Chassis

Course Objectives:

1. To impart knowledge of importance of maintenance at regular intervals.
2. To impart adequate knowledge of maintenance and maintenance methods required in modern service station.
3. To impart knowledge of dismantling of various parts/ assemblies and check the parts for proper functioning.
4. To impart knowledge of proper maintenance of the components results in good fuel economy, least environmental pollution and reliability.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
01	**Auto Workshop Layout & Equipments** 1.1 General safety precautions and procedures. 1.2 Equipment used in automobile repairing (Specification, Working, Application,): Shop-cutters, Pullers, Stud-extractor, Torque wrench, Piston ring expander, Piston ring groove cleaner, Wheel Balancer, Wheel Aligner, Arbor Press, Drill Press, Tyre Changer, Car Washer, Battery Charger, Battery Tester, Valve Grinder, Honing Machine, Cylinder Boring Machine, Pneumatic tools, lubrication equipment. 1.3 Measuring and Testing Equipments: Feeler gauge, Cylinder bore gauge, Compression gauge, Ignition timing tester, Spark plug tester, tyre inflator gauge, Micrometer, Callipers, Protectors and their maintenance. 1.4 Vehicle Service Equipment: Air Compressor, Oil Sprayer, Lubricators, Voltage current and resistor tester, Coil condenser tester, Tachometer, Exhaust Gas Analyzer, Diesel Smoke meter. 1.5 Lifts and Hoists: two post lift, four post lift, scissor type lift. 1.6 Service Station and Types. 1.7 Criteria for site selection of Service Station. 1.8 Layout of Modern Auto workshop.	10
02	**Maintenance Management and Record Keeping** 2.1 Necessity of maintenance 2.2 Types of maintenance (Breakdown, Preventive and Predictive) systems, their applications and comparison,. 2.3 Total Productive Maintenance (concept only) 2.4 Vehicle maintenance schedules: Daily, weekly, monthly &	04

		periodic maintenance. for various vehicles Replacement schedule of consumable items (like lubricants, coolant, Filters etc.) 2.5 General servicing procedure. Decision to repair or replace. 2.6 Workshop documents and records - Job Card. 2.7 Operations and Procedures - Workshop activities, Manpower requirements and criteria, billing and warranty claim procedure, insurance claim procedure, customer feedback taking procedure, handling of customer complaints, study of customer service Index (CSI).	
03	**Part 1** 3.1	**Engine Maintenance** 3.1.1 Checking and Servicing of following engine components: Cylinder head, Cylinder block, Cylinder liners, Piston, Piston Ring, Crank-shaft, Connecting rod, Valves etc. 3.1.2 Troubleshooting in: Cylinder head, Cylinder block, Cylinder liners, Piston, Piston Ring, Crank-shaft, Connecting rod, Valves etc.	06
	Part 2 3.2	**Engine System Maintenance** 3.2.1 Fuel feed system: Injector cleaning and testing, FIP phasing and calibration. 3.2.2 Lubrication system service: Change of oil filter, Check oil pump and diagnose causes for excessive oil consumption, external oil leakage and low oil pressure in an automobile engine. 3.2.3 Maintenance of cooling systems and its components: water pump, radiator, thermostat - anticorrosion and antifreeze additives. Bleeding and Flushing of cooling system. 3.2.4 Troubleshooting of Ignition System service and Electronic Ignition System. 3.2.5 Troubleshooting of Engine fuel system, Cooling system, Lubrication system.	08
04	**Part 1** 4.1	**Chassis Maintenance** 4.1.1 Troubleshooting of Clutch. 4.1.2 Troubleshooting of manual Gearbox. 4.1.3 Troubleshooting of Propeller shaft, Rear axle and Differential system. 4.1.4 Troubleshooting of hydraulic brakes, Bleeding of hydraulic brakes and parking brake adjustment. 4.1.5 Troubleshooting of Dependent and Independent suspension systems, Lubrication of leaf springs.	08
	Part 2 4.2	**Chassis frame, Steering, Wheels, Tyre and Body Maintenance** 4.2.1 Symptoms and possible faults in steering system. 4.2.2 Power steering diagnosis and troubleshooting. Fluid flushing, bleeding air out and pressure testing.	08

		4.2.3 Procedure of wheel alignment and procedure of wheel balancing. 4.2.4 Care of wheels and tyres, tyre rotation. 4.2.5 Defects in vehicle chassis and body - diamond, banana, twist, mash type damages. 4.2.6 Body repairs- denting, denting tools and equipments (jacks) 4.2.7 Chassis alignment equipment. 4.2.8 Repainting procedure, patch work. 4.2.9 Painting defects. 4.2.10 Adjustment of doors and locks.	
Total Hours			**42 Hrs**

Weightage distribution in both objective, short and broad answer type questions:

Group	Module Number	Weightage (%)
A	1 & 2	30
B	3	35
C	4	35

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Understand the use of relevant tools and equipments required in garage practices.
CO2	Understand different types of maintenance methods/ techniques for vehicles.
CO3	Identify different problems associated with different types of engine systems.
CO4	Interpret troubles associated with transmission system of a vehicle.
CO5	Recognize various difficulties associated with steering, wheels, braking and suspension systems.

Text Books:

1. Vehicle Maintenance and Garage Practice, Dhruv U. Panchal, Jayesh P. Maniar and Jigar A. Doshi, PHI Learning.
2. Automobile Maintenance Service and Repair, Mohd Saad Saleem and Saimah Khan, Bluerose Publishers Pvt. Ltd.
3. Dr. Kripal Singh, Automobile Engineering (Vol. – 1), Standard Publishers Distributors, New Delhi.

Reference Books:

1. S. Srinivasan, Automotive Mechanics, Tata McGraw Hill Education, New Delhi.
2. S.K. Gupta, Automobile Engineering, S.Chand, New Delhi.
3. R.B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi.

Proposed Syllabus of Modern Vehicle Technology

Course Code	AEPE611
	Modern Vehicle Technology
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	Automotive Engine, Automotive Chassis, Automotive Powertrain

Course Objectives:

1. Impart knowledge on the applications and working of different transducers, sensors & actuators used in automobiles.
2. Impart knowledge on the uses of modern peripheral systems in automobiles.
3. Impart knowledge on advanced technologies related to IC Engine.
4. Impart knowledge on advanced safety equipment used in automobiles.
5. Impart knowledge on different modern features of automobiles.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
01	**Applications of Transducers, Sensors & Actuators** 1.1 Concept of general measurement system & difference between Mechanical and electrical/ electronic instruments; 1.2 Measurement of Temperature: Working of Thermocouple and Thermister; 1.3 Measurement of Speed: Contact less electrical tachometer, Inductive, Capacitive type tachometer, Stroboscope; 1.4 Measurement of Force: Strain gauge load cell 1.5 Electrical method for moisture measurement; 1.6 Electromechanical Type Transducer – Potentiometric resistance type, Inductive, Capacitive, Piezometric; Photoelectric. 1.7 Basic requirement of Sensors, Functions, Applications and Circuitry arrangement of various Sensors such as Mass Air flow rate sensor, Exhaust gas Oxygen concentration, Throttle plate angular position, Crankshaft angular position, Coolant temperature, Intake air temperature, Manifold absolute pressure (MAP), Vehicle speed Sensor. Transmission gear selector position, Methanol sensor, Rain Sensor & Rain sensing wiper. 1.8 Working Principal and Functions of various Actuators such as Solenoid Actuators, Motorized Actuators, and Stepper motors.	10

02	**Modernization in IC Engine Performance** 2.1. Concept of Fuel Injection System in SI Engine 2.2. Necessity of Fuel Injection System 2.3. Working principle and function of various fuel injection system in SI engine (MPFI, TBI & GDI) 2.4. Advantages of Fuel Injection System w.r.t Pollution 2.5. Concept of Common Rail Direct Injection System 2.6. Necessity of CRDI System 2.7. Basic concept of single cylinder pump, Inline pump, Distributor or Rotary pump, Low pressure pump & High-Pressure Pump 2.8. Working principle and function of CRDI System 2.9. Advantages of CRDI System w.r.t Pollution	12
03	**Modernization in Peripheral systems** 3.1. Security Systems. Remote keyless entry, Anti-theft system, Alarm system. 3.2. Entertainment and peripheral systems. Integrated communications, Proximity sensors	04
04	**Advance Safety Equipment** 4.1. Seat Belts, Seat Belts pre-tensioners, Smart seatbelt reminder. 4.2. Concepts of Crash test, Crash sensors. 4.3. Air bags - Introduction of air bags, Dual stage air bags, Side Airbags. 4.4. Tyre pressure monitoring system 4.5. Pedestrian Protection & Night vision with pedestrian detection.	06
05	**Modern Features in Automobile** 5.1. Power Sliding doors. 5.2. Electronic stability/ Skid-control system, Traction control system. 5.3. Advanced Driver Assistance System (ADAS), Adaptive cruise Control System (ACC) 5.4. Hill Hold Assist (HHA) 5.5. Telescopic steering wheel/ adjustable pedals. 5.6. Rear mounted Radar & Cameras. 5.7. Electromagnetic suspension and levitation. 5.8. Automatic Lift Axle. 5.9. Regenerative Braking Systems. 5.10. Continuous Variable Transmission. 5.11. Intelligent Parking Assist System, Self-Parking	10
Total Hours		**42 Hrs**

Weightage distribution in both objective, short and broad answer type questions:

Group	Module Number	Weightage (%)
A	1 & 2	40
B	3 & 4	30
C	5	30

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Describe construction, functions and applications of various sensors and actuators used in modern vehicle.
CO2	Identify and describe the advanced features of IC Engine
CO3	Identify and describe various advanced peripheral system used in automobile.
CO4	Demonstrate various safety features and equipment used in modern vehicle.
CO5	Identify various modern features for better functioning of vehicle.

Text Books:

1. S.K. Gupta, Automobile Engineering, S.Chand, New Delhi.
2. R.B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi.
3. A.K. Babu, S.C. Sharma, T.R. Banga, Automobile Mechanics, Khanna Publishers

Reference Books:

1. S. Srinivasan, Automotive Mechanics, Tata McGraw Hill Education, New Delhi.
2. Dr. N.K. Giri, Automobile Technology, Khanna Publishers

Proposed Syllabus of Vehicle Aerodynamics & Design

Course Code	AEPE 612
Course Title	Vehicle Aerodynamics & Design
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PE
Prerequisites	Fluid Mechanics

Course Objectives:

1. To understand basic components of Aerodynamics.
2. To understand different aerodynamic effects on a vehicle.
3. To understand the ergonomic of vehicle
4. To understand the optimization process of aerodynamics drag and noise
5. To know about wind tunnel testing
6. To understand the different type vehicle body structure and material

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1.	**Introduction of Aerodynamics:** **1.1.** Road Resistance- Air Resistance – Gradient Resistance Aerodynamic force – Drag force – Lift force – side force - effect of Aerodynamics force in car body **1.2.** Aerodynamics - pitching – yawing - Rolling moment **1.3.** Cross wind sensitivity – Recirculating flow - diffusers **1.4.** Definition of Aerofoil- vehicle as a bluff body - Mechanics of air flow **1.5.** Pressure distribution on a vehicle. **1.6.** Concept of visibility - **1.7.** Concept of blind sport **1.8.** vehicle ergonomic – importance – application – different ergonomic in different vehicle	**10**
2	**Directional Stability:** 2.1. Concept of aerodynamic stability 2.2. Distribution of weight -for three wheeled vehicle and four wheeled vehicle 2.3. Driving with trailer 2.4. Stability of vehicle on slope (derivation & numerical problems). 2.5. Stability of vehicle on turns (derivation & numerical problems).	**6**
3.	**Optimization of vehicle body design:** 3.1. Side wind problems- Dirt accumulation on the vehicle 3.2. Wind noise – different type of noise – minimization process of wind noise in vehicle 3.3. Front and rear wind shield angle – Boat tailing – Hatch back, fast back and square back 3.4. Dust flow patterns at the rear – Effect of gap configuration – Effect of fasteners 3.5. Various body optimization techniques for minimising drag	**8**

4	**Wind Tunnels testing and other testing method** **4.1.** Wind Tunnels for Automotive Aerodynamics Introduction – Principles of wind tunnel technology – Full scale wind tunnels - scale model testing **4.2.** Component balance to measure forces and moments– Stress with scale models — Measurement techniques – Equipment and transducers **4.3.** Road testing methods – Numerical methods –advantages – Limitation-application.	**6**
5	**Vehicle structure** **5.1.** Bus Body Details – Types - mini bus - single decker, two level, split level and articulated bus. **5.2.** Bus Body Lay Out: Floor height, engine location, entrance and exit location, seating dimensions. **5.3.** Constructional details - Frame construction, Double skin construction-Types of metal section used-Regulations-Conventional and Integral type construction. **5.4.** Commercial Vehicle Details - Types of body - Flat platform, drop side, fixed side, tipper body, tanker body. **5.5.** Light commercial vehicle body types, Dimensions of driver's seat in relation to controls, driver's cabin design. **5.6.** Body Materials, Trim And Mechanisms: Steel, timber, plastics, GRP, properties of materials-Corrosion anticorrosion methods, selection of paint and painting process, body trim items.	**12**
	Total Hours	**42**

Weightage distribution in both objective, short and broad answer type questions:

Group	Module Number	Weightage (%)
A	1 & 2	45
B	3 &4	25
C	5	30

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Identify the different aerodynamics forces on a vehicle.
CO2	Estimate the vehicle stability against aerodynamics force and slope.
CO3	Differentiate the various drag testing method.
CO4	Identify the different types of wind noise in a vehicle.
CO5	Classify different vehicle structure.

Text book:

1. Aerodynamics of Road vehicles, W. H. Hucho, Butterworths Co. Ltd., 1997
2. Vehicle Body Engineering, A.K. Babu, Publisher: Khanna Publisging House, 2021
3. Vehicle Body Engineering & Dynamics, Bheemasen S Korlahalli, Vinayak S Naik, Eastern Book Promoters Belgaum, 2022.

References

1. Automotive Aerodynamics: Update SP-706, SAE, 1987.
2. Vehicle Aerodynamics, SP-1145, SAE, 1996.

Proposed Syllabus of Vehicle Body Engineering

Course Code	AEPE613
	Vehicle Body Engineering
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	PC
Prerequisites	Engineering Mechanics and Automotive Chassis

Course Objectives:

1. Impart knowledge on different car body assemblies, design and safety aspects.
2. Impart knowledge on different bus body assemblies, design and safety aspects.
3. Impart knowledge on different commercial vehicle body assemblies and design aspects.
4. Impart knowledge on aerodynamics approach of automobiles.
5. Impart knowledge on different body materials.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
01	**Car Body Details** 1.1 Car body assembly: underbody, upper body, closure (brief ideas only) 1.2 Types of Car body: Saloon, Hatchback, Convertible, Coupe, Estate Car, Limousine, Racing and sports car (brief ideas with diagram and example) 1.3 Visibility: Types, Regulations, Improvements in visibility, parameters influencing visibility, visibility tests 1.4 Different methods of improving space in cars: lowering floor level, position of spare wheel, compact design 1.5 Seat design: Driver seat design, passenger seat design 1.6 Car body construction: Requirements, methods of construction, various panels in car bodies 1.7 Safety aspect of car body design, Safety requirements for car (seat belt, ABS, TCS, Vehicle cruise control, Electronic stability control, Tyre pressure monitoring system, rear view camera, Supplementary restraint system)	10
02	**Bus Body Details** 2.1. Types of Bus body: based on capacity, based on distance travelled, based on construction, based on comfort, shape and style, engine location. 2.2. Types of material used: crowns, angles and flanges, sections	08

	2.3. Regulation wise overall dimensions 2.4. Constructional details: Components of body structure, design of integral bus, sequence of bus building operation, double skin construction, safety aspect 2.5. Seat design: Driver's seat (with geometrical relations), passenger seat	
03	**Commercial Vehicle Details** 3.1. Different vehicle bodies: Based on engine mounting, based on load-carrying platform, based on their functions, purposes, based on driver cab design 3.2. Flat platform bodies: Trailer, Tipper, Tanker (different types according to type of liquid carrying, shape of the tanker and bulkheads) 3.3. Dimensions of driver's seat in relation to controls, driver's cab design, regulations.	06
04	**Vehicle Aerodynamics** 4.1. Introduction and importance of vehicle aerodynamics 4.2. Aerodynamics forces and moments: Drag, cross wind force, lift. Rolling, pitching, yawing moments (definition and mathematical expression only, no derivation) 4.3. Aerodynamic drag and types: Profile, induced, friction, interference, cooling and ventilation drag 4.4. Drag co-efficient, aerodynamic lift and Pitching moment ((definition and mathematical expression only) 4.5. Side wind effects on forces and moments: side wind on lift force, side wind on moments 4.6. Various body optimization techniques for minimum drag 4.7. Wind tunnels, types of wind tunnels, wind tunnel testing (brief idea)	12
05	**Body Materials** 5.1. Materials used in body construction: steel sheet, aluminum alloy, timber, plastics, GRP, Textiles, Glass, Rubbers (uses, properties, pros and cons) 5.2. Body trim materials: Exterior and Interior trim 5.3. Body mechanisms: Window winder, windshield wiper, windshield washer, door lock mechanism, keyless entry.	06
Total Hours		**42 Hrs**

Weightage distribution in both objective, short and broad answer type questions:

Group	Module Number	Weightage (%)
A	1 & 2	30
B	3 & 4	50

C	5	20

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Classify vehicle body according to body shape
CO2	Illustrate the different types and components of car body.
CO3	Illustrate the different types and components of bus and commercial body
CO4	Explain the concept, importance and testing of aerodynamics in vehicle body design.
CO5	Explain different vehicle body materials with their merits and demerits.

Text Books:

1. A.K. Babu, Vehicle Body Engineering, Khanna Publishers
2. J. Powloski, Vehicle Body Engineering, Business Books Ltd, London

Reference Books:

1. J.G. Giles, Body Construction and Design, Butterworth and Co.
2. Dr. N.K. Giri, Automobile Technology, Khanna Publishers

Proposed Syllabus of Garage Practice Lab

Course Code	AEPC 602
	Garage Practice Lab
Number of Credits and L-T-P	1 [L – 0, T – 0, P - 2]
Course Category	PC
Prerequisites	Automotive Engines and Automotive Chassis

Course Objectives:

1. To impart knowledge service procedure in Automobile Workshop.
2. To impart adequate knowledge and skills of overhaul and reconditioning of various engine components.
3. To impart knowledge of overhaul and testing of Injector and FIP of diesel engine for proper functioning.
4. To impart knowledge of overhaul different chassis components for proper functioning of vehicle.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
01	**Study General service procedure in Authorize workshop** 1.1 Four wheeler service, 1.2 Two wheeler service.	02
02	**Overhaul and reconditioning procedures of multi-cylinder petrol/Diesel engine.** 2.1 Cylinder head for warpage and cracks, Refacing by grinding or cutting, Straightening cylinder heads. 2.2 Cylinder block for measurement of ovality and taperedness, Cylinder boring, Honing process, Changing of liners. 2.3 Piston and piston rings for wear, appearance, Piston head for signs of deposits and detonation, oversize piston, ring groove clearance, removing and refitting rings. 2.4 Valve refacing in valve refacer machine, Valve Seat cutting, setting and grinding to match with valves. Lapping of Valves. 2.5 Inspection of Crank Shaft, Assessment of workability and determination of undersize condition of journals. Setting procedure of Crank Shaft of Multi cylinder Engines in Crank Shaft regrinding machine for grinding both crank pin and main Journals, Check for eccentricity of cranks. 2.6 Dismantling and assembling of fuel injectors of a petrol engine, identifying components and refitting. 2.7 Removing the radiator from vehicle, checking it for leak, repairing, flushing, cleaning the radiator and refitting. Removing the thermostat valve, checking and refitting. 2.8 Checking of lubrication system – change oil filter, check oil pump, diagnose causes for excessive oil consumption, external oil leakage, and low oil pressure in an automobile engine.	12

03	**Overhaul and Testing** 3.1 Overhaul and testing of diesel fuel injector, 3.2 Overhaul and testing of single and multi- cylinder fuel injection pumps. 3.3 Calibration, phasing, and spray tests.	06
04	**Overhaul, dismantling and assembling procedures of different chassis components:** 4.1 Leaf springs, coil springs, torsion bar & Telescopic Shock absorber and McPherson strut. 4.2 Wheel Balancing: - Static and Dynamic. 4.3 Wheel alignment – Mechanical and Electronic method	08
Total Hours		**28 Hrs**

Note: Module 01, 03 & 04 is compulsory and minimum 04 job has to be done from module 02.

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Illustrate the servicing procedure in Automobile workshop.
CO2	Interpret maintenance methods/ techniques of multi-cylinder petrol/Diesel engine.
CO3	Recognize various difficulties associated with diesel fuel injection system.
CO4	Recognize various difficulties associated with suspension and wheel.

Text Books:

1. Vehicle Maintenance and Garage Practice, Dhruv U. Panchal, Jayesh P. Maniar and Jigar A. Doshi, PHI Learning.
2. Automobile Maintenance Service and Repair, Mohd Saad Saleem and Saimah Khan, Bluerose Publishers Pvt. Ltd.
3. Dr. Kirpal Singh, Automobile Engineering (Vol. – 1), Standard Publishers Distributors, New Delhi.

Reference Books:

1. S. Srinivasan, Automotive Mechanics, Tata McGraw Hill Education, New Delhi.
2. S.K. Gupta, Automobile Engineering, S.Chand, New Delhi.
3. R.B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi.

Proposed Syllabus of Major Project-II

Course Code	PR602
Course Title	Major Project-II
Number of Credits and L-T-P	6 [L – 0, T – 0, P - 6]
Course Category	Major Project
Prerequisites	All Subject Related to Automobile Engineering

Course objectives: -
After completing this course, the students will be able

- To understand the solution process of real life problem
- To achieve the potentiality of doing team work
- To understand the gap between academic knowledge and actual real life problem solving knowledge.
- To prepare the project repot in a skillful way.

Project group:-

1. Formation of project group: Maximum 8 to 12 students per batch.
2. Each project group should select work by consulting the guide.

Course Content:-
It is the *continuation of major project –I* (which is stared on Semester V) during this major project-II, the project work have to be completed by each group.

After completion of the project, each students group should prepare *one* comprehensive report to indicate what are **observed, learnt and contribution** during the project work. The student may contact guide for completing the project and prepared the final report on the assigned topic. The project report should be signed by the guide and HOD of concerned department.

The format of the project report will be as following

Sl. No.	Particulates
1	Title page
2.	Deceleration page
3	Acknowledgement
4	Certificate from guide
5	Abstract
6	Objective
7	Literature review/ background survey/history
8	Present work
9	Methodology
10	Observation
11	Conclusion
12	Future Scope
12	References

Internal Assessment:

1. Project Report: The project Report will be evaluated on the basis of following criteria (as applicable) **40 Marks**

Sl. No.	Criteria for evaluation of Project.
1	Originality
2	Adequacy and purposeful write up
3	Organizations, format, drawing, sketches, style, language
4	Practical applications and relationships with basic theory
5	Concept taught in the outcome

2. Attendance and work process **20 Marks**

External Assessment:

Seminars must be arranged for the student based on the project report, in presence of project guide, Internal Committee constituted by the concerned department of the Institute and External examiner/s. The evaluation will be based on the following criteria; **40 Marks**

Sl. No.	Criteria for evaluation of Report
1	Quality of content presented
2	Proper Planning for presentation
3	Effectiveness of presentation
4	Depth of knowledge and skills
5	Viva voce

Course Outcome:
At the end of the course, the student will be able to:

CO1	learn new skills and supplement knowledge
CO2	Learn & practice communication and teamwork skills.
CO3	Learn strategies like time management, multi-tasking, real time technical knowledge etc.
CO4	Can apply their knowledge to generate new idea for doing some application oriented work.
CO5	Gain practical experience in a real working environment.

Proposed Syllabus of Driving Practice Lab

Course Code	AEPE603
	Driving Practice Lab
Number of Credits and L-T-P	2 [L – 0, T – 0, P - 2]
Course Category	PC
Prerequisites	None

Course Objectives:

1. To develop knowledge of behavior of driver.
2. To impart adequate knowledge of vehicles control and its responses.
3. To impart knowledge of Traffic signs and traffic rules to follow as per government.
4. To develop knowledge and skills of vehicle driving.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
01	**Driving Theory:** **1.1 Qualities required for a good driver**: Good behavior, Patience, Responsibility, Self Confidence, Anticipation, Concentration, Courtesy, Consideration for other Road Users, Defensive Driving, Knowledge of Vehicle Controls, Maintenance, simple mechanism and effect of Competitiveness, **1.2** Over-Confidence and impatience, avoid road rage. **1.3** Legitimate dress code for driving. 1.4 Right of way, yielding. 1.5 Concept of blind spot.	02
02	**2.1 Know Your Vehicle**: Simple introduction to Automobile Engines and their working. Location of VIN plate / Engine serial Number. On Board Diagnosis position, Several Air Bags position, Dual power mode driving, First Aid Kit, Jacking point. **2.2 Vehicle Controls and its Response**: 2.1 Foot Control – Accelerator, Brake, Clutch 2.2 Hand Control – Steering Wheel, Gear Shifting Lever, Hand Brake, Switches for direction Indicators, Head Lights, Horns, Ignition Switch, Dim-Bright Switch, and Wiper switch – Speed Control etc. 2.3 Other Controls – Rear View mirrors – Types- adjustments, Audio controls, Air Condition controls, Instrument Cluster, Description of Speedo meter/Trip meter, RPM Gauge, Oil pressure Gauge, Temp Gauge, Fuel Gauge, and Ammeter other Gauges and Dials, Tail lamps, Audio warnings, Wind Screen – their location and operation. 2.4 Adjustment of seat and seat belts, door locks, Spare key location.	04
03	2.1 **Traffic signs:** Mandatory signs, Cautionary signs, Informatory	02

	signs, Traffic signals displayed on roads. 2.2 Study of Traffic Rules as per State & central Government specification.	
04	**Driving Practice—** **4.1 Pre – Driving Checks**: Before sitting on driver seat, After sitting on driver seat Checking the document pertains to the driver and vehicle. **4.2 Starting**: Precautions and Procedure to be followed while starting. Clutch down start. Accelerator: Proper use of Accelerator. Moving: Precautions to be followed while moving. Use of first gear. **4.3 Clutch Practice**: Biting and Balance point. **4.4 Steering practice**: In basic yard - In bare land and between tyres to develop Judgment Straight, Left and Right Turn, Steering in '8', shaped bend. **4.4 Gear Practice**: Selection of gears, up shifting, down shifting. **4.5 Brake practice**: In level ground and in slope. **4.6 Driving on the Road** **4.7 Overtaking Practice**: Overtaking stationary and Moving vehicles from Left and Right Side. **4.8 Three Point Turn, Five Point Turn, 'U' Turn Practice** **4.9 Reverse Practice.** **4.10 Parking Practice.** **4.11 Other Maneuvers related to driving etc.** **4.12 Demonstrate safe, responsible and proper parking techniques and etc.**	20
Total Hours		**28 Hrs**

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Explain the responsible driving attitudes.
CO2	Recognize various controls of vehicles.
CO3	Demonstrate the traffic signs and traffic rules
CO4	Develop competence in integrating the attitudes, skills and knowledge of safety and driving responsibilities into the correct execution of motor skills in traffic.

Note: Experiment no. 04 can be pursued by using suitable Driving Training Simulator.

Proposed Syllabus of Seminar

Course Code	SE602
Course Title	Seminar
Number of Credits and L-T-P	2 [L – 0, T – 0, P - 4]
Course Category	Seminar
Prerequisites	Up to 5th semester all subject knowledge

Course Objectives:

1. Identify and compare technical and practical issues related to the area of course specialization.
2. Outline annotated bibliography of research demonstrating scholarly skills.
3. Prepare a well-organized report employing elements of technical writing and critical thinking.
4. Demonstrate the ability to describe, interpret and analyze technical issues and develop competence in presenting.

Guideline

Each student has to select a recent topic of latest technology in the area of Automobile Engineering and present a seminar in front of all students of the class. He/ She has to prepare a PowerPoint presentation of the selected topic of minimum 10 slides are the total presentation will be approximately 10 minutes duration. There will be interactive session between the presenter and rest of the students including the faculty members of the dept at the end of presentation. A student has to present at least 2 nos. of seminar during a semester and to submit the report for evaluation.

The Seminar Report will be evaluated on the basis of following criteria (as applicable)

60 Marks

Sl. No.	Criteria for evaluation of Seminar Report
1	Originality
2	Adequacy and purposeful write up
3	Organizations, format, drawing, sketches, style, language
4	Practical applications and relationships with basic theory
5	Concept taught in the course outcome
6	Practical applications, relationships with basic theory and concept taught in the course
7	Attendance record, daily diary, quality of Internship Report.

Seminars must be arranged for the student based on his/her training report, before an Internal Committee constituted by the concerned department of the Institute. The evaluation will be based on the following criteria:

40 Marks

Sl. No.	Criteria for evaluation of Seminar Report
1	Quality of content presented
2	Proper Planning for presentation
3	Effectiveness of presentation
4	Depth of knowledge and skills
5	Viva voce

Course Outcome:

At the end of the course, the student will be able to:

CO1	Establish motivation for any topic of interest and develop a thought process for technical presentation.
CO2	Organize a detailed literature survey and build a document with respect to technical publications.
CO3	Analysis and comprehension of proof-of-concept and related data.
CO4	Effective presentation and improve soft skills.
CO5	Make use of new and recent technology for creating technical reports

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)

Syllabus
of

Compulsory Subjects & Open Electives as per New Syllabus [OE]

Part-III (6th Semester)

2023

List of Compulsory Subjects & Open Electives as per New Syllabus 2023

Compulsory Subject		
Subject Code	**Name of the Subject**	**Page Number**
HS 302	Entrepreneurship & Start-ups	3-6
Open Elective - I		
Subject Code	**Name of the Subject**	**Page Number**
OE 302	Engineering Economics & Project Management	7-9
Open Elective - II		
Subject Code	**Name of the Subject**	**Page Number**
OE 304/1	Electric Vehicle Technology	10-12
OE 304/2	Industrial Management	13-16
OE 304/3	Industrial Safety	17-22
OE 304/4	Disaster Management	23-25
OE 304/5	Environmental Science & Engineering	26-29
OE 304/6	Renewable Energy	30-32
OE 304/7	Mechatronics	33-35
OE 304/8	Internet of Things	36-37
OE 304/9	Sustainable Development	38-40
OE 304/10	Medical Electronics	41-42
OE 304/11	Occupational Health & Safety Engineering	43-49
OE 304/12	Industrial Hazards & Modern Waste Management	50-54
OE 304/13	Export & Import Management	55-60
OE 304/14	Industrial Management and Safety	61-66
OE 304/15	Electrical Machines & Control	67-73
OE 304/16	Artificial Intelligence	74-75
OE 304/17	Operations Research	76-77
OE 304/18	Soft Computing Techniques	78-79
OE 304/19	Construction Management	80-82
OE 304/20	Solid Waste Management	83-85
OE 304/21	Sustainable Architecture	86-87
OE 304/22	Machine Learning	88-90
OE 304/23	Web Designing	91-94
OE 304/24	Energy & Environment Control in Metallurgical Industries	95-97
OE 304/25	Network Security Management and Administration	98-101
OE 304/26	Internet of Things (for CFS)	102-103
OE 304/27	Network Security Management and Administration Lab (for CFS)	104-105
OE 304/28	Internet of Things Lab (for CFS)	106
OE 304/29	Economic Policies in India	107-108

Course Title	**Entrepreneurship and Start-ups**
Course Code	HS 302
Number of Credits	3
Pre Requisites	None
Total Contact Hours	3(L: 2; T: 1)/week = 45 hrs
Course Category	HS

Course Learning Objectives

1. To raise awareness, knowledge and understanding of enterprise/ entrepreneurship.
2. To motivate and inspire students toward an entrepreneurial career.
3. To understand venture creation process and to develop generic entrepreneurial competences.
4. To introduce students to the basic steps required for planning, starting and running a business.
5. To familiarise students with the different exit strategies available to entrepreneurs.

Course Outcomes:

After completing the course students will able to:

CO 1	Identify qualities of entrepreneurs, develop awareness about entrepreneurial skill and mindset and express knowledge about the suitable forms of ownership for small business
CO 2	Comprehend the basics of Business idea, Business plan, Feasibility Study report, Project Report and Project Proposal
CO 3	Understand the concept of start-up business and recognise its challenges within legal framework and compliance issues related to business.
CO 4	Make a Growth Plan and pitch it to all stakeholders and compare the various sources of funds available for start-up businesses

Detailed Course Content

Unit	**Name of the Topic**	**Hours**
1.	**ENTREPRENEURSHIP – INTRODUCTION AND PROCESS** • Concept, Competencies, Functions and Risks of entrepreneurship • Entrepreneurial Values& Attitudes and Skills • Mindset of an employee/manager and an entrepreneur • Types of Ownership for Small Businesses o Sole proprietorship o Partnerships o Joint Stock company- public limited and private limited	10

	companies • Difference between entrepreneur and Intrapreneur	
2.	**PREPARATION FOR ENTREPRENEURIAL VENTURES** • Business Idea- Concept, Characteristics of a Promising Business Idea, Uniqueness of the product or service and its competitive advantage over peers. • Feasibility Study – Concept – Locational, Economic, Technical and Environmental Feasibility. Structure and Contents of a standard Feasibility Study Report • Business Plan – Concept, rationale for developing a Business Plan, Structure and Contents of a typical Business Plan • Project Report- Concept, its features and components • Basic components of Financial Statements- Revenue, Expenses (Revenue & capital exp), Gross Profit, Net Profit, Asset, Liability, Cash Flow, working capital, Inventory. Funding Methods-Equity or Debt. Students are just expected to know about the features and key inclusions under, Business Plan and Project Report. <u>They **may not** be asked to prepare a Business Plan/ Project Report/ Project Feasibility Report in the End of Semester Examination.</u>	20
3.	**ESTABLISHING SMALL ENTERPRISES** • Legal Requirements and Compliances needed for establishing a New Unit- o NOC from Local body o Registration of business in DIC o Statutory license or clearance o Tax compliances	03
4.	**START-UP VENTURES** • Concept & Features • Mobilisation of resources by start-ups: Financial, Human, Intellectual and Physical • Problems and challenges faced by start-ups. • Start-up Ventures in India – Contemporary Success Stories and Case Studies to be discussed in the class. Case studies have been included in the syllabus to motivate and inspire students toward an entrepreneurial career from the success stories. <u>No questions are to be set from the case studies.</u>	04

5.	**FINANCING START-UP VENTURES IN INDIA** • Communication of Ideas to potential investors – Investor Pitch • Equity Funding, Debt funding – by Angel Investors, Venture Capital Funds, Bank loans to start-ups • Govt Initiatives including incubation centre to boost start-up ventures • MSME Registration for Start-ups –its benefits	06
6.	**EXIT STRATEGIES FOR ENTREPRENEURS** • Merger and acquisition exit, Initial Public Offering (IPO), Liquidation, Bankruptcy – **Basic Concept only**	02

Examination Scheme

- **End Semester Examination: 60 marks**

  Suggested Question Paper Scheme for End Semester Examination

  Group A: 20marks

Question Type	Number of questions to be set	Number of questions to be answered
MCQ, Fill in the blanks, True or False (Carrying 1 mark each)	25	20

  Group B: 40marks

Question Type	Number of questions to be set	Number of questions to be answered
Subjective Type questions (Carrying 8 marks each)	10	5

- **Internal Assessment: 40 marks**
  - Class test : 20 marks
  - Assignment: 10 marks
  - Class attendance: 10 marks

Suggested Learning Resources

Sl. No.	Title of Book	Author	Publication
1.	Entrepreneurship Development	Sangeeta Sharma	Prentice Hall of IndiaLearning Private Ltd
2.	Entrepreneurship Development	S. Anil Kumar	New Age International
3.	Fundamentals of Entrepreneurship	Sangram Keshari Mohanty	Prentice Hall of India Learning Private Ltd
4.	Fundamentals of Entrepreneurship	Dr. G.K. Varshney	Sahitya Bhawan Publication
5.	Managing New Ventures: Concepts and Caseson Entrepreneurship	Anjan Raichaudhuri	Prentice Hall of India Learning Private Ltd
6.	How to Start a Business in India	Simon Daniel	Buuks, Chennai
7.	Entrepreneurship and Small Business Management	S.S. Khanka	S. Chand & Sons, New Delhi
8.	Entrepreneurship Development and Business Ethics	Abhik Kumar Mukherjee & Shaunak Roy	Oxford University Press
9.	Entrepreneurship Development and Business Ethics	Dr B Chandra & Dr B Biswas	Tee Dee Publications
10.	Entrepreneurship Development Small Business Entrepreneurship	Poornima Charantimath	Pearson Education India

Draft Syllabus of Engineering Economics & Project Management

Course Code:	OE301
Course Title:	Engineering Economics & Project Management
No. of Credits:	3 (L:3, T:0,P: 0)
Prerequisites:	NIL
Course Category:	Open Elective (Compulsory for all branches)

Course Objectives:

- To acquire knowledge of basic economics to facilitate the process of economic decision making.
- To acquire knowledge on basic financial management aspects.
- To develop the idea of project plan, from defining and confirming the project goals andobjectives, identifying tasks and how goals will be achieved.
- To develop an understanding of key project management skills and strategies.

Group-A

Unit-I (INTRODUCTION, THEORY OF DEMAND & SUPPLY) [9 hours]

Introduction to Engineering Economics, the relationship between Engineering and Economics

Resources, scarcity of resources, and efficient utilization of resources.

Opportunity cost, rationality costs, and benefits

Theory of Demand: the law of demand, different types of demand, determinants of demand, demand function, price elasticity of demand.

Theory of Supply: determinants of supply, supply function.

Market mechanism: Equilibrium, basic comparative static analysis (Numerical problems)

Unit-II (THEORY OF PRODUCTION & COSTS) [10 hours]

Concept of production (goods & services), Different factors of production (fixed and variable factors), Short-run Production function (Graphical illustration), and Long run production function (returns to scale).

Theory of Cost: Short-run and long-run cost curves with graphical illustration, basic concept on total cost, fixed cost, variable cost, marginal cost, average cost etc.

Economic concept of profit, profit maximization (numerical problems)

UNIT-III (DIFFERENT TYPES OF MARKET AND ROLE OF GOVERNMENT) [4 hours]

Perfect Competition: Features of Perfectly Competitive Market.

Imperfect Competition: Monopoly, Monopolistic Competition, and Oligopoly.

Role of government in Socialist, Capitalist and Mixed Economy structure with example.

Group-B

Unit-I (CONCEPT OF PROJECT) [4 hours]

Definition and classification of projects)

Importance of Project Management.

Project life Cycle [Conceptualization→Planning→Execution→Termination]

Unit-II (FEASIBILITY ANALYSIS OF A PROJECT) [10 hours]

Economic and Market analysis.

Financial analysis: Basic techniques in capital budgeting– Payback period method, Net Present Value method, Internal Rate of Return method.

Environmental Impact study–adverse impact of the project on the environment.

Project risk and uncertainty: Technical, economical, socio-political, and environmental risks.

Evaluation of the financial health of a project–Understanding the basic concept of Fixed & Working Capital, Debt & Equity, Shares, Debentures etc., and different financial ratios like Liquidity Ratios, Activity Ratios, Debt-equity ratio & Profitability Ratio (Basic concept only).

N.B: Knowledge of financial statements is not required; for the estimation of ratios the values of the relevant variables will be provided.

Unit-III (PROJECT ADMINISTRATION) [8 hours]

Gantt Chart– a system of bar charts for scheduling and reporting the progress of a project (basicconcept).

Concept of Project Evaluation and Review Technique (PERT) and Critical Path method (CPM):basic concept and application with real-life examples.

ExaminationScheme:

A. SemesterExaminationpatternof60 marks:

1. Objective type Question (MCQ, Fill in the blanks, and Very Short question-1 mark each): At least five questions from each unit. [Total marks: 20]

2. Subjective questions: Eight questions to be answered taking at least three from each group. (Two questions should be given from each unit). [Total marks: 40]

B. Assignment (10Marks) Guide line

for Assignment (10 Marks)

Students should be instructed to prepare a report on a project (preferably the Major Project they prepare in 6th Semester), using a popular project management software in IT/ Computer Laboratory, under the guidance of the Lecturer in Computer Science & Technology and Lecturer in Humanities.

C. Class Test: Two examinations 20 marks each. Take best of two.

D. Attendance: 10 Marks

Suggested reference books:

1 *Principles of Economics – Case and Fair, Pearson Education Publication*
2 *Principles of Economics – Mankiw, Cengage Learning*
3 *Project planning, analysis, selection, implementation and review – Prasannachandra–Tata McGraw Hill.*
4 *Project Management – Gopala Krishnan – Mcmillan India Ltd*

Proposed Syllabus for Electric Vehicle Technology

Course Code	OE II
Course Title	Electric Vehicle Technology
Number of Credits and L-T-P	3 [L – 2, T – 1, P - 0]
Course Category	PC
Prerequisites	Basic Electrical and Electronics Engineering Mechanics Strength of Materials

Course objectives: -

After completing this course, the students will be able

- To understand the basics of electric vehicle history and components.
- To understand properties of batteries.
- To understand the electrical machine properties and classifications.
- To understand the properties of electric vehicle drive systems
- To understand the concepts of hybrid electric vehicles.

Unit	Contents	No of Classes
1	**Introduction** History of Hybrid and Electric Vehicles; Social and Environmental importance of Hybrid and Electric Vehicles; Basic Components, Vehicle mechanics: Roadway fundamentals, Vehicle kinetics, Dynamics of vehicle motion; Propulsion System Design.	6
2	**Battery:** Introduction to Energy Storage – Cell and Battery fundamentals, Battery capacity, Open circuit voltage and Terminal voltage, Charge/Discharge rate, State of charge/discharge, Depth of discharge, Battery energy density & Specific energy, Battery power density & Specific power, Battery efficiency. Different traction batteries – Lead Acid battery, Nickel based batteries, Sodium based batteries and Lithium based batteries – Li- ion& Li-poly. Battery Management System – Definition, Parts: Power Module, Battery, DC/DC Converter, Load, Communication channel, Battery Pack Safety Troubleshooting of EV Batteries.	10

3	**DC Electrical Machines:** Basic concept of DC Electric motors, Types, Power & torque generation, Power-flow diagram and braking in motors. DC Motors Basic Construction & Principle of Operation, Brushless &Brushed DC Motor Torque equation and Torque-speed characteristic of a DC motor Regenerative braking of a DC motor 3.2.3 Permanent Magnet Brushless DC Motor(Basic concept only) 3.3 Advantages & disadvantages of DC Electrical Machines 3.4.Applications in connection to EVs 3.5 Troubleshooting of DC Electrical Machines	8
4	**AC Electrical Machines:** Three phase AC Induction Motor Basic Construction & Principle of Operation Synchronous speed, Slip, Simplified Torque expression andTorque-slip Characteristics Per-phase equivalent circuit of an Induction Motor Speed control methods Regenerative braking of an Induction motor Permanent Magnet Synchronous Motor(Basic concept only) Switched Reluctance motor(Basic concept only) Advantages & disadvantages of AC Electrical Machines Applications in connection to EVs Troubleshooting of AC Electrical Machines	8
5	**Electric Vehicle Drive Train:** 5.1 Transmission configuration; Components: Gears, Differential, Clutch, Brakes; Regenerative braking, Motor sizing; Fuel efficiency analysis.	4
6	**Hybrid Electric Vehicles:** Types: Parallel, Series, Parallel and Series configurations; Drive train; Sizing of components; Basics of Micro, Mild, Mini, Plug-in and Fully hybrid.	6
	Total Classes	**42**

Weight age distribution

Group Name	Unit Number	Weightage (%)
A	1	10
	2	20
B	3	25
	4	25
C	5	10
	6	10

Course Outcome:

At the end of the course, the student will be able to:

CO1	Understand the basics of electrical vehicle history and components.
CO2	To understand properties of batteries as energy storage device for Electric Vehicles.
CO3	To understand working and properties of different electrical motors in connection to their applications in Electric Vehicles.
CO4	Understand the properties of electrical vehicle drive systems.
CO5	Understand the concepts of hybrid electric vehicles.

Reference Books:

1. Electric & Hybrid Vehicles – A.K. Babu, Khanna Publishing House, New Delhi, 2018
2. Electric & Hybrid Vehicles – Design Fundamentals - Iqbal Hussain, Second Edition, CRC Press, 2011.
3. Electric Vehicle Technology Explained - James Larminie, John Wiley & Sons, 2003.
4. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals - MehrdadEhsani, YiminGao, AliEmadi, CRC Press, 2010.
5. Electric Vehicle Battery Systems - Sandeep Dhameja, Newnes, 2000.
6. Electric and hybrid vehicles technologies, modeling and control - amirkhajepour / saberfallah / avestagoodarzi

e-References

https://www.primecom.tech/blogs/news/what-types-of-motors-are-used-in-electric-vehicles